Integrated Carbon Capture and Utilization

From Flue Gas to Polymer: Carbon Capture Integrated with Methanol-to-Monomer Production

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1 Project Charter

Objective:

To design a process that captures post-combustion Carbon Dioxide (CO_2) from a steel refinery, then converts it to methanol through catalytic hydrogenation, raising methanol into light olefins through a Honeywell-UOP based MTO system modeling an economic, environmental, technologic performance of the proposed pathway.

Deliverables:

- Block frow diagram and preliminary PFDs
- Mass and energy balances
- Equipment descriptions and operating conditions
- Environmental assessment
- Technological/economical evaluation
- Gantt chart
- Final report and presentation

Criteria for Successful Project:

- Achieve 830,000 tons/yr olefins with 99.9% purity
- Evaluate 45Q credit impacts and CO_2 reduction
- Showcase an efficient cost model versus using fossil-based ethylene propylene
- Explore feasible alternative processes to Honeywell-UOP MTO process for future business.

Project Scope:

- CO_2 capture from steel refinery's flue gas (20% CO_2 , 25% CO, 3% H_2, and impurities)
- Integration of Hydrogen (SMR + capture/external supply)
- Methanol synthesis over $ZnO/Cu/Al_2O_3$
- MTO processing with SAPO-34 method
- Olefin purification to polymer grade
- Alternative polymer production pathways from methanol
- Market, environmental, safety, and economic analysis

2 Executive Summary

The project plans for a pathway of carbon utilization that converts post-combustion CO_2 from a oil refinery burn stacks located in Galveston Bay, Texas to valuable light olefins (ethylene and propylene) using an integrated CO_2 to methanol to olefins process. The system proposes to address both challenges of industrial decarbonization and a growing demand for polymers by changing wasteful gas to a key petrochemical feedstock.

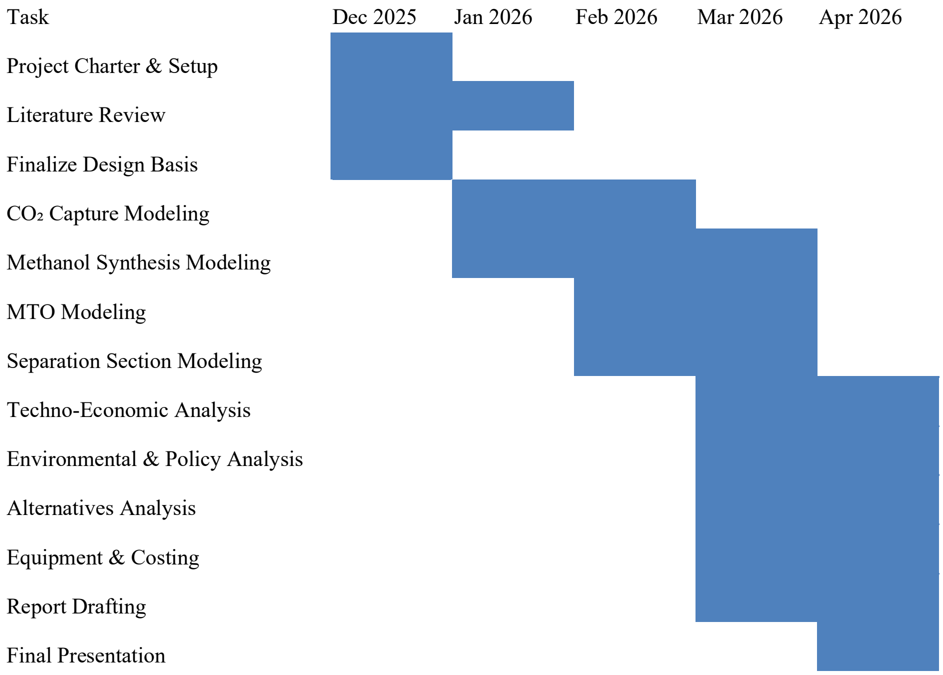
The capture is purified and then reacted with hydrogen over a catalyst of ZnO/Cu/ Al_2O_3 to produce a high-purity methanol. This methanol is then converted in a Honeywell-UOP-style methanol-to-olefins (MTO) reactor with the use of a SAPO-34 catalyst system to produce 830,000 tons per year of olefins (469,000 tons per year of ethylene and 361,000 tons per year propylene). The design basis of the project assumes 8000 operating hours per year and 20 years of plant operations.

Considering the economic structure of the project, the process utilizes the value of methanol upgrading and U.S. potential 45q tax credits for CO_2 usage. Environmentally, the project proposes a system that reduces reliance on fossil-based steam cracking and provides sustainable olefin production. Technologically, both MTO and CO_2 to methanol are processes with significant amounts of industrial research and development.

The project explores the MTO configuration and analyzes other alternative methanol-to-polymer routes including methanol-to-polymer (MTP), methanol-to-gas (MTG) with oxidating coupling variants, olefin recovery, and direct methanol-to-polyolefin ideas. These alternative processes are seen as comparisons for long term economic strength and scalability.

The final design demonstrates that CO_2 methanol is a feasible platform chemical for sustainable polymer production, with the provision of a hydrogen source and optimized catalyst management. The integrated system places carbon capture as a valuable chemical manufacturing method.

3 Gantt Chart



4 Introduction

4.1 Problem Definition and Significance

Global demand for light olefins, primarily ethylene and propylene, is steadily increasing due to their crucial role as precursor building blocks in the production of polymers, fibers, chemicals, and consumer materials. The International Energy Agency (IEA) reports that olefin consumption continues to rise at ~3–4% annually, making petrochemicals one of the fastest-growing sources of oil demand worldwide.

Traditionally, ethylene and propylene are produced through steam cracking of hydrocarbons derived from crude oil and natural gas. This conventional method is highly energy-intensive, environmentally damaging, and responsible for substantial CO_2 emissions and industrial waste. For example, ethylene production alone releases approximately 1.5–3.0 tonnes of CO_2 per tonne of product, depending on feedstock and energy source.

These fossil-based processes contribute heavily to global greenhouse gas emissions, resource depletion, and worsening climate impacts. The chemical industry now accounts for ~18% of global industrial CO_2 emissions, with light olefin production being a major contributor.

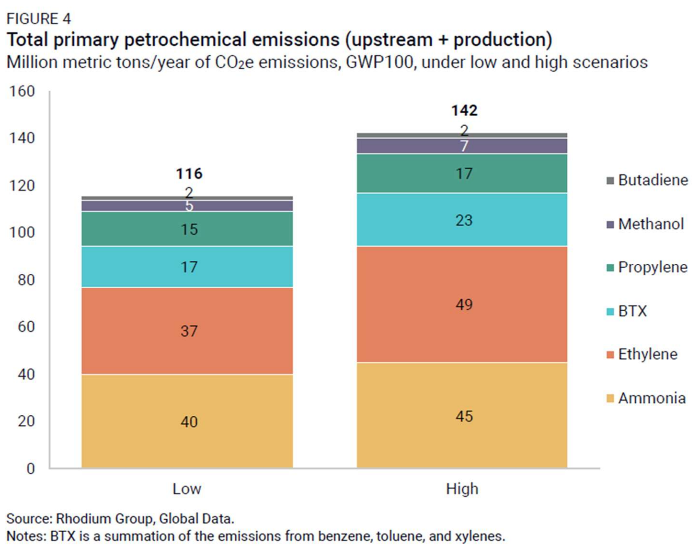


Figure 1: *Total Primary Petrochemical Emissions (Bower, 2025)*

To address this issue, there is a growing focus on closing the carbon loop. Rather than releasing CO_2 during olefin production, captured CO_2 can be utilized as the carbon feedstock, providing a green alternative that replaces petrochemical extraction and mitigates emissions from existing industrial activity.

4.2 Carbon Capture Motivation

Carbon Capture, Utilization, and Storage (CCUS) is recognized as a critical climate mitigation pathway by global scientific and policy organizations. Industrial flue gas, especially from energy-intensive manufacturing such as oil refining, contains large volumes of CO_2 with concentrations that make capture economically viable. Oil Refining alone contributes ~13–15% of global CO_2 emissions.

Galveston Bay, Texas is home to one of the largest integrated oil refineries in the world, producing millions of tonnes of flue-gas CO_2 annually. This provides a convenient and continuous source of carbon for upgrading into valuable chemical products. In 2024, **MGBR produced 8.2 million tons of GHG**. This accounted for approximately 0.5% of total GHG emissions in the world. Being able to reduce and use this source to create polymers can contribute to the creation of a more sustainable world. (Fagelson, Nathaniel. 2024)

By integrating carbon capture with chemical conversion technologies such as CO_2 hydrogenation and methanol-to-olefins (MTO) synthesis, previously emitted carbon can now be preserved within a circular process. This reduces direct emissions from oil refinery operations, dependence on fossil hydrocarbons, net global atmospheric carbon intensity

The approach not only prevents waste emissions but transforms CO_2 into a revenue-generating feedstock, aligning environmental sustainability with industrial profitability.

4.3 Importance of Sustainable Polymer Production

The shift toward sustainable polymer manufacturing is a major transformative objective across chemical and consumer product industries. Traditional plastics rely on finite fossil resources and create long-lived waste streams. By contrast, sustainable polymer frameworks prioritize Carbon-neutral feedstocks, advanced catalysis to lower energy consumption and improve selectivity, circular-economy recycling methods, and low-emission energy integration across chemical plants

Organizations such as the National Academies and IEA note that integrating CO_2 utilization into chemical value chains could reduce global carbon emissions by billions of tonnes annually while supporting economic growth.

Sustainable polymer production, lowers GHG emissions across the entire product lifecycle, reduces reliance on crude-oil-derived hydrocarbons, encourages innovation in lightweight, high-performance materials, and supports global decarbonization legislation and ESG goals

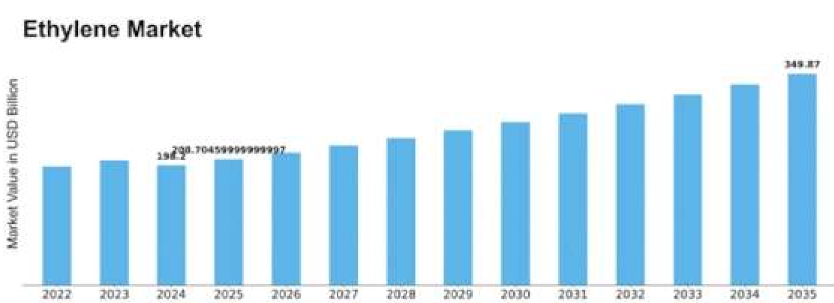
By converting CO_2 into monomers such as ethylene and propylene, materials already fully compatible with existing polymer infrastructure, this approach accelerates the transition toward a carbon-circular plastics economy without requiring redesign of consumer products.

5 Market Analysis

5.1 Global Ethylene & Propylene Market

The global polymer industry is one of the most significant segments of the chemical value chain, encompassing the packaging, construction, automotive, textile, and consumer goods. Recent market estimates project the global market value to be USD 790-800 billion in 2024, forecast to reach USD ~1.34 trillion by 2034. Corresponding to a compound annual growth rate (CAGR) of 5-5.5% over the next decade (Research, Polymers Market Strengthen industrial application with bio-based materials, lightweight thermoplastics, and Asia-Pacific scale, 2025).

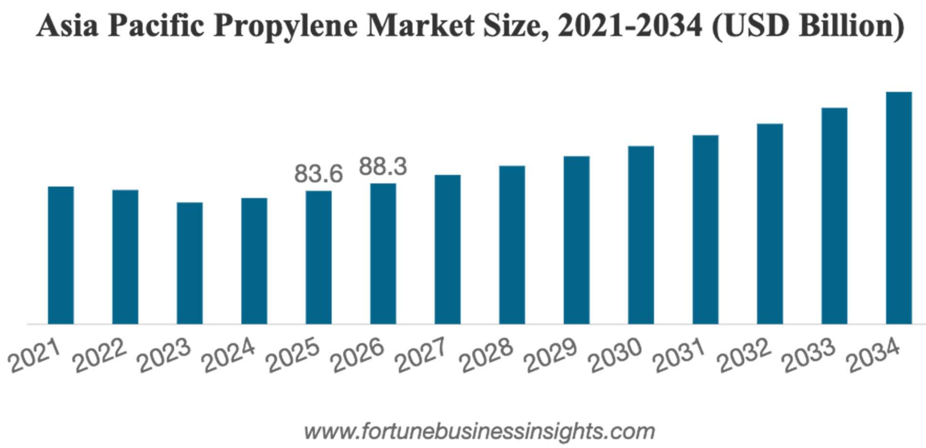
Pertaining to this process, as per Market Research Future analysis, the market value of ethylene is estimated at USD 198.2 billion in 2024. The ethylene industry is forecasted to grow from USD 208.71 billion in 2025 to USD 349.87 billion by 2035 demonstrating a CAGR of 5.3% during the forecast period of 2025-2035. (Market, 2024)



The ethylene market is going through a phase of advancements and evolving demand patterns. Ethylene has been increasingly used in various applications like automotive, packaging, and construction are significant drivers of growth. As companies shift towards more sustainable practices and reduce their carbon footprint, the ethylene market will likely encounter innovations and enhance efficiency while reducing environmental impact. North America remains the largest market for ethylene due to its demands in the plastic sector. However, Asia-Pacific is the fastest growing region which is powered by growing urbanization and industrialization (Market, 2024).

The global propylene market size was valued at USD 134.3 billion in 2025 and is projected to grow from USD 141.1 billion in 2026 then USD 216.2 billion by 2034 at a CAGR 5.4% during the forecast period. Unlike ethylene, the Asia-Pacific region dominated the market with a share of 62.2% in 2025 (Fortune Business Insights, 2026).

Propylene is widely used in the automot**ive, constructive,** and packaging industries due to its importance in the manufacturing of a wide range of commonly utilized goods. Despite the pandemic effecting the demand due to less manufacturing activity, the product remained critical for medical supplies. However, the market is predicted to recover with the revival of global demand (Fortune Business Insights, 2026).



In terms of region, the Asia Pacific dominates the market, accounting for 49.95% of the market share in 2024. Specifically, China's olefin market is expected to be driven by the domestic demand, expansion of the production capacity, and the material's diverse applications. In contrast, polymer demand in Europe is relatively mature and currently constrained by high energy costs and regional restrictions imposed by petrochemical companies. North America shows moderate growth, supported by low-cost ethane feedstock and export-oriented investments in PE and PP capacities.

Polymer Markets remain cyclical, with profitability heavily influenced by crude oil, natural gas, and NGL feedstock prices, as well as over-capacity cycles in steam cracking and PDH units. Recent analyses have highlighted an oversupply situation since 2022, which has been pressuring margins and prompting plant closures or rationalization in higher-cost regions. Nevertheless, long-term projections still indicate positive demand growth for both commodity and engineering plastics, with value shifting toward specialty-grade, high-performance materials and applications that replace heavier or more energy-intensive materials.

A significant trend is the transition towards more sustainable polymers and circular plastics. Markets for "sustainable" or circular plastics, including recycled, bio-based, and CO2-derived polymers, are expected to grow significantly faster than conventional plastics, with a predicted CGR value of >9-13% over the next decade (Research, Commodity Plastics Market Size to Worth USD 666.76 Billion by 2034, 2025). Specific drivers for this market's growth include the EU's mandate for recycled content targets in packaging, producer responsibility schemes, bans and taxes on single-use plastics, and corporate net-zero commitments from major FMCG, automotive, and retail companies. These trends do not eliminate the need for ethylene and propylene in the near future, but instead shift the preferred source and carbon footprint of the monomers. Technologies that enable low-carbon or circular feedstock routes are therefore increasingly attractive as part of decarbonization strategies for polymer value chains.

Within this context, an 830 ktma plant represents a small volume in the global polyolefin market, but it is significant on a regional scale. Furthermore, the proposed plant addresses both the growing demand for polyolefins and the pressure to reduce life-cycle greenhouse-gas emissions associated with conventional monomer production. Overall, the global polymer market is large and growing, and as a result, it is becoming increasingly carbon-constrained. This drives the need for a more robust foundation and strategic route for developing a CO2-based methanol-to-olefin process, which will provide a lower-carbon source of polymers.

5.2 Value Chain Overview

The value chain for this project spans the transformation of captured carbon dioxide into a high-value polymer. This pathway leverages clean energy, catalytic conversion technologies, and established downstream polymer markets to create both economic and environmental value. The proposed plan begins with CO_2 as a feedstock, proceeds through methanol synthesis, and then to a methanol-to-olefin (MTO) conversion, ultimately producing light olefins. At each stage, value is added, but costs and carbon-intensity risks also accumulate. More importantly, the ultimate competitiveness and sustainability of the chain depend on two key drivers: the cost and carbon footprint of the hydrogen used, as well as the availability of policy incentives.

CO_2, captured from methan combustion burn stack flue gas, is purified and fed into a hydrogenation loop where H_2 and CO_2 react to produce methanol. The hydrogen is supplied either externally or generated on-site through electrolysis or reforming with carbon capture. This methanol serves as an intermediate, a liquid, transportable, and storable chemical that can substitute for conventional fossil-derived methanol or feedstocks. From methanol, the core MTO reactor converts the methanol into light olefins (ethylene, propylene) plus some byproducts. These olefins serve as the foundational building blocks for polyolefin elastomers, solvents, and numerous petrochemical derivatives. Thus, the chain bridges waste or excess CO_2 to high-value commodity chemicals, effectively recycling carbon. The marketing of this product will focus on this aspect, as well as its strong sustainability positioning. By leveraging CO_2 utilization and a lower-carbon footprint, buyer demands for environmentally improved materials are satisfied. Certification, emissions disclosure, and compliance with circular economy targets become strategic market differentiators. Customers are incentivized to adopt these materials to meet corporate ESG goals and comply with regulatory mandates regarding the use of recycled or low-carbon plastics.

However, the value of this chain is susceptible to the hydrogen source and its carbon intensity. Suppose hydrogen is produced by conventional steam-methane reforming without effectively producing grey hydrogen. In that case, the upstream CO2 reduction may be negated by the CO2 emitted during hydrogen production, undermining any climate advantage and increasing the net carbon footprint. Suppose hydrogen is supplied via blue hydrogen or green hydrogen. In that case, the carbon intensity of the overall olefin output drops significantly, and the overall value of the process becomes negative, consequently resulting in a loss of money due to the high cost of these types of hydrogen. The most significant solution to this would be to produce the grey hydrogen in plant, while simultaneously capturing the carbon. However, the logistics of this need to be researched to implement. The decision of whether to purchase hydrogen or integrate on-site hydrogen production further influences operating costs and plant autonomy.

The economics of this chain are also heavily dependent on the government and regulatory incentives around carbon capture and utilization. In a U.S. context, the tax-credit framework under Section 45Q offers financial incentives per metric ton of CO_2 captured and either stored or utilized (Jones, 2023). Under Section 45Q, a facility capturing CO_2 and utilizing it in chemical conversion may qualify for the utilization credit rate (IRS, 2025). Section 45Q tax credits provide up to \$60 per metric ton of CO_2 utilized and up to \$130 per metric ton stored permanently, depending on the pathway and compliance requirements, according to congressional analysis. This incentive may significantly improve the economics of upstream carbon capture, provided the facility meets the eligibility criteria.

A crucial consideration in this value chain is that monomers alone do not maximize profitability. Ethylene and propylene are globally traded commodities with volatile margins and are most economically valuable when converted into higher-value derivatives, such as polymers. Polyethylene and polypropylene command significantly higher selling prices and more stable demand profiles than their monomer precursors; therefore, a strategic integration of downstream polymerization units is necessary to capture the value of this conversion process fully. Without access to polymerization, the project would remain exposed to monomer price cyclicality and face reduced returns on investment.

Given these dependencies, the value of this project hinges on the costs of hydrogen sourcing, carbon intensity, final product form, and the policy environment. With green or blue hydrogen and a favorable tax or credit regime, such as 45Q, CO_2-derived olefins can become both economically competitive and environmentally attractive. Without them, the process may struggle to compete with conventional, well-optimized steam-cracking-based olefin production, especially in regions with low-cost fossil feedstocks. This value chain demonstrates how the transformation of low-value CO_2 into polymer materials represents a complete industrial upgrade pathway. Each step adds value, from waste carbon to commodity chemicals to multipurpose manufactured materials, while also aligning with policy and regulatory incentives that support decarbonization. Achieving the greatest value capture requires integrated polymerization capacity, rather than selling monomers alone, ensuring that the economic benefits of the conversion process are fully realized at the highest-value point in the product chain.

5.3 Pricing of Raw Materials

The selection, cost, and sourcing strategy of raw materials represent critical drivers of both the economic and environmental performance of the proposed CO_2-to-polymers facility. The major inputs to this process include captured carbon from flue gas, hydrogen, water, and supporting chemicals.

Methanol is both a potential feedstock and a reference commodity. As of late 2025, contract methanol price indicates a typical methanol value around USD 320 per tonne in the North American region, and USD 250-290 per tonne in the Asian Middle Eastern Region (Price-Watch, 2025). Given these market numbers, the raw-material cost baseline is approximately USD 300–350/tonne, depending on the region, freight, and purity requirements. This baseline helps contextualize the competitiveness of methanol produced from CO_2 conversion within internal operations.

Hydrogen is a critical raw material that directly influences the cost and carbon footprint of the entire chain. Current hydrogen prices depend heavily on production method: grey, blue, and green. Grey hydrogen is from natural gases and carbon capture. It remains the most cost-effective option, with recent market prices ranging from USD 0.96 to USD 3 per kilogram. Blue hydrogen is a fossil-driven form of hydrogen that also utilizes carbon capture. It has a higher cost, typically around USD 2-3.5 per kilogram, reflecting the purification costs. Ultimately, green hydrogen is produced through water electrolysis using renewable energy sources. It is currently the most expensive, with typical market values ranging from USD 3.5 to USD 6.0 per kilogram. Due to the high hydrogen demand per tonne of methanol for the CO_2 to methanol process, the choice of hydrogen supply method becomes a major lever in the project. From this, the scenario of using grey hydrogen in combination with a circular CO_2 capture method becomes the most profitable.

Flue gas is not purchased as a raw material but instead can act as a potential profit source. As previously discussed, tax credits are available for up to $60 per metric ton of CO_2 utilized. It is crucial in this process to be accredited by the government and to be subject to its regulation to take advantage of this potential source of profit.

Water is required for hydrogen production, cooling, and diluent/reactant in various units. In many industrial contexts, the cost of raw water is relatively low compared to the costs of fuel and hydrogen; however, in some geographies, water scarcity may make water a non-trivial operating expense. For rough economic modeling, water can often be assumed at a modest tariff or treated as an internal utility cost. Utility costs (electricity for H_2, steam, cooling water, and compression) should be treated separately in the energy balance and utility section, rather than as raw materials per se.

Other chemicals, such as catalysts, also contribute to operating costs but are best handled separately under reagents and operating consumables rather than raw material costs for bulk inputs.

Table 1: Raw Materials Pricing Table

Raw Material	**Pricing Basis**	**Price Range**	**Notes**
Grey Hydrogen	USD/kg	0.96-3.0	Low cost but highest carbon intensity
Blue Hydrogen	USD/kg	2.0-3.5	Includes cost of carbon capture and removal infrastructure
Green Hydrogen	USD/kg	3.5-6.0	Cleanest option
Flue Gas	-	-	Offset by tax credits
Water	USD/m^3	1.00	Specific to Gary, IN
Amine Solvent	USD/kg	1.0-2.0	Consumed gradually over operating life
SAPO-34 Catalyst	USD/kg	10-50	Periodic replacement
$Cu/ZnO/Al_2O_3$ Catalyst	USD/kg	20-80	Periodic replacement

5.4 Pricing of Products

The economic viability of the MTO-based plant depends not just on the raw material costs but the market value of the output products. This process produces two main products, ethylene and propylene, as well as numerous byproducts that can be sold for profit.

As of late 2025, the cost of ethylene in the NA region is USD 0.56-0.6 per kilogram. Propylene prices are assessed to be USD 0.83-0.84 per kilogram. The jump from monomer to resin adds considerable margin, provides direct access to high-demand resin markets, and is likely essential to realize the full economic potential of the plant.

Table 2: Pricing of Products Table

Product	Pricing Basis	Price Range	Notes
Ethylene	USD/tonne	560-600	NA 2025 price
Propylene	USD/tonne	830-840	NA 2025 price
Fuel Gas (Methane)	USD/tonne	120-160	
Ethane	USD/tonne	150-220	Assuming low grade
C_3^+ Heavier Hydrocarbons	USD/tonne	220	Sold as mixed NGL/fuel-range stream

6 Judging Criteria

Due to rising global CO_2 emissions, up 0.8% in 2024, reaching an all-time high of 37.8 Gt CO_2, there is a strong incentive to develop alternative and more sustainable industrial processes. Much of this increase stems from industrial activity, which has contributed to a 50% rise in atmospheric CO_2 concentrations (ppm) since 1750**.**

Currently, the combustion of natural gas, liquid fuels, and coal remains the primary driver of long-term carbon emission growth. Countries such as China, Japan, Indonesia, India, and the United States are among the largest contributors due to their substantial energy consumption. However, it is worth noting that the United States and Japan have seen a steady decline in CO_2 emissions since the early 2000s.

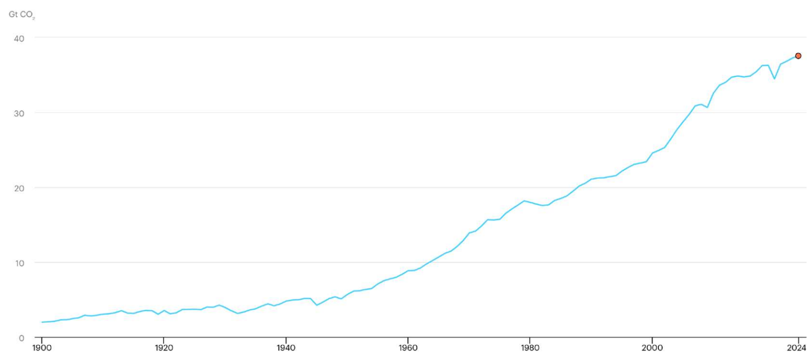


Figure 2. *Global CO_2 Emissions Trend (1900-2024), (IEA, 2025).*

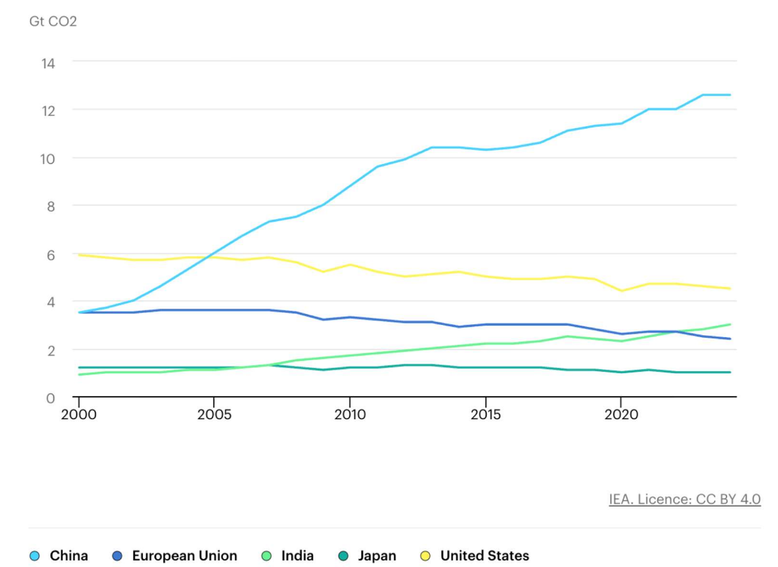


Figure 3. *CO_2 Total Emissions by Region (2000-2024), (IEA, 2025).*

For our Methanol-to-Olefins (MTO) project, we considered the significant environmental impacts of carbon emissions. Our process begins by capturing CO_2 and converting it to methanol (MeOH) through hydrogenation. Rather than relying on conventional processes that generate additional flue gas through the combustion of natural gas or coal, our design instead utilizes burn stack flue gas from methane combustion in oil refineries, from which CO_2 is purified and recovered.

Through heat integration and process optimization, the CO_2 emissions associated with MeOH production are reduced from 3.15×10^{-1} $tCO_2/tMeOH$ to 6.77×10^{-3} $tCO_2/tMeOH$, representing roughly a 98% decrease (see figure below). This reduction is supported by high process selectivity, including a CO_2 conversion of approximately 99%, enabled by an optimized CO_2:H_2 ratio (1:7) and an industrially relevant hydrogenation catalyst.

Downstream, the produced MeOH is converted into light olefins using a proprietary SAPO-34 catalyst. During this reaction, the catalyst gradually accumulates solid carbon ("coke"). To maintain catalyst performance, SAPO-34 undergoes regeneration by controlled combustion, which produces additional CO_2. In our design, this CO_2 is captured and recycled back into the MeOH synthesis loop, further lowering net emissions while also providing a modest increase in overall plant profitability.

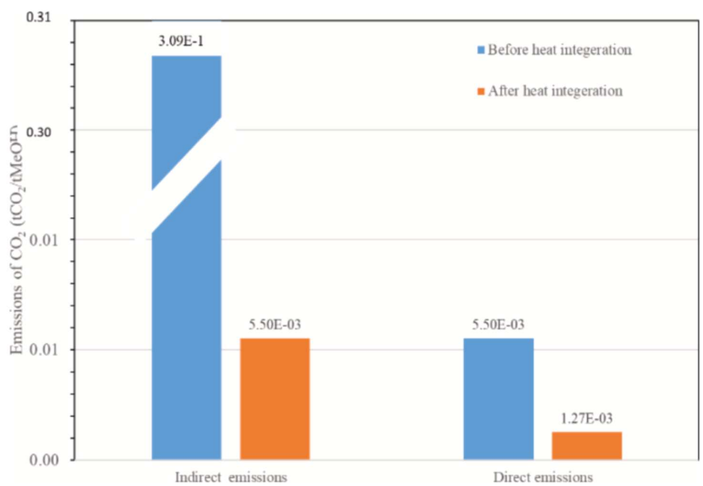


Figure 4. *Total Estimated CO_2 Emissions for the Hydrogenation Process Before and After Heat Integration (Yusuf & Almomani, 2023).*

Our project also prioritizes efficient energy use. As mentioned earlier, the initial step of our process involves capturing CO_2 from steel-mill off-gases using an amine-based absorbent solution. Conventional amine absorption is highly energy intensive because the absorption of CO_2 is an exothermic process, while regenerating the absorbent is endothermic, requiring substantial cooling and heat input. This makes CO_2 capture one of the major operating costs in typical plants. To reduce this energy burden, our process incorporates a mixture of substituted amines that enables the absorbent solution to spontaneously form two liquid phases during CO_2 absorption: a CO_2-rich phase and a CO_2-lean phase. The CO_2-rich phase is denser, and the two phases become immiscible, allowing for separation by gravity. Only the CO_2-rich phase is sent to the regenerator, significantly lowering the required heat duty. In contrast, traditional single-phase absorbent systems send the entire absorbent stream to regeneration, leading to much higher energy consumption.

Other areas of our overall process demonstrate energy efficiency as well. In the MTO section, the fluidized-bed reactor that converts MeOH to light olefins maintains its temperature through heat supplied by the catalyst regenerator. The regenerator is directly connected to the reactor to enable catalyst recirculation, and when air reacts exothermically with the SAPO-34 catalyst during regeneration, the released heat is transferred directly into the reactor. This reduces the external energy required to maintain the optimal reactor temperature for achieving the desired selectivity and conversion.

A final example of energy efficiency in our project is found in the CO_2-to-MeOH synthesis section. Using Aspen's Energy Analyzer v11, a heat-exchanger network (HEN) is developed that significantly improved energy integration and reduced utility consumption. By strategically incorporating three heat exchangers into the final design, overall energy savings increased by 63.2%. External hot-utility demand decreased from 3.13×10^{1} to 1.4×10^{-2} GW (a 99.6% reduction), and external cooling-utility demand decreased from 3.62×10^{1} to 2.11×10^{1} GW (a 41.7% reduction).

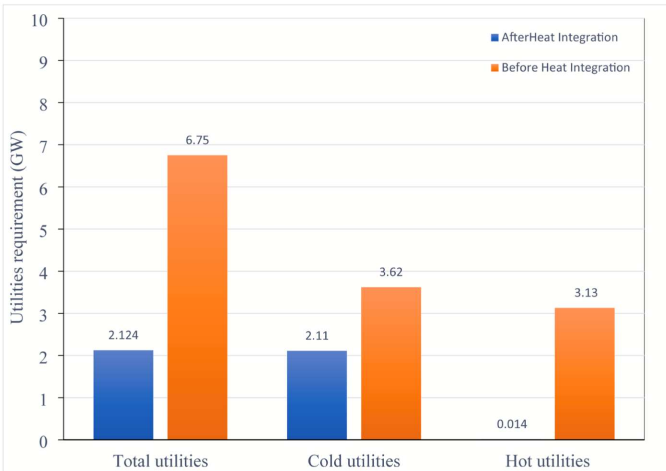


Figure 5. *Required External Utility Before and After the Optimized H.E.N (Yusuf & Almomani, 2023)*

One additional factor that strengthens the appeal of our project is the availability of federal carbon-management incentives. Although Indiana does not impose a state-level carbon tax on industrial facilities, the federal 45Q tax credit and provisions within the Inflation Reduction Act (IRA) apply to qualifying operations. These programs offer substantial financial incentives for facilities engaged in carbon capture or utilization, including those located in Indiana. In summary, the potential for federal tax credits, the environmental advantages of our process compared to similar technologies, and the efficient use of energy collectively make our project a strong and viable endeavor.

7 Process Description

7.1 Chosen Process

This process converts carbon dioxide into high-value olefins through a fully integrated system consisting of CO_2 capture, methanol synthesis, and methanol-to-olefins (MTO) conversion. Flue gas containing CO_2 is first treated to remove unwanted molecules before entering an amine absorption unit, where CO_2 is selectively captured and purified for downstream use. The CO_2 and H_2 gas are fed into a catalytic hydrogenation reactor operating over $Cu/ZnO/Al_2O_3$ catalyst to produce crude methanol. This mixture is cooled and flashed to remove unreacted gases before entering a purification chain, where high-purity methanol is recovered via distillation.

The purified methanol stream is then introduced into the methanol-to-olefins (MTO) reactor, where it undergoes dehydration and catalytic conversion to a mixture primarily composed of ethylene and propylene, with other byproducts such as CO_2. Downstream separation is achieved through a series of absorbers and cryogenic distillation towers. Propylene and ethylene are recovered with high purity as monomers.

In an ideal configuration, the process is expanded to create a fully integrated, carbon-circular manufacturing route to polymers. In this enhanced design, Hydrogen demand is met via on-site steam-methane reforming (SMR) or renewable-powered electrolysis. SMR-based operation includes a secondary CO_2 capture step, ensuring that nearly all carbon is retained in the system rather than emitted.

The captured CO_2 and generated hydrogen are then converted to methanol through catalytic hydrogenation in the same reactor loop used in the current design. The methanol product is dehydrated and oligomerized in an MTO reactor to form light olefins. These olefins are further processed through cryogenic separation into high-purity ethylene and propylene, just as in the current configuration. However, unlike the existing design, these monomers are not sold directly. Instead, they are immediately fed to Ziegler–Natta-based polymerization reactors to produce polyethylene and polypropylene pellets, enabling significant product value uplift and supply-chain stability.

Additionally, off-gas streams containing residual hydrogen, methane, and carbon dioxide are recycled. Hydrogen is recovered for reuse, methane is directed to reforming, and carbon dioxide is returned to the capture section, forming a closed-loop carbon cycle. Water generated during methanol synthesis and MTO conversion is treated and recycled to reduce freshwater consumption.

This idealized process configuration enhances environmental and economic outcomes by maximizing carbon utilization efficiency, minimizing purchasing of external raw materials, increasing product value through polymer production, and enabling access to U.S. tax credits such as the federal 45Q CO_2 utilization incentive.

The following subsections provide a comprehensive explanation of the process, reaction pathways, and operating conditions involved in each major processing operation.

7.1.1 CO_2 Capture Process

7.1.1.1 Description

The selected process for CO2 carbon capture for the methanol to monomer synthesis is a patent focused on utilizing flue gas from burning fuels. The patent is focused on a simple absorption to separation extraction of flue gas for CO2 separation to be used in further synthesizing. The absorption is used with an Amine separation and catalyst. This allows for the separation of clean gas that can be released into the air. While a mixture of CO2 and water where it will be fed into a series of separators and knock out drums to remove CO2 and Water. The water will be disposed of, and the CO_2 will be compressed for the development of methanol. An in-detail analysis of the chosen process starts with the feeding of flue gas from the burn stack into a absorption (Column T-100) for clean gas of N_2 and O_2 to be split from CO_2 and H_2O. Once CO_2 and H_20 are separated from the flue gas both will be added into their respective pressures of 115 Celsius and 1 bar to be added into a Stripping column for the separation of CO_2, H_2O and remaining impurities for the final step of splitting with a series of know out drums to remove remain water and produce a 98% pure CO_2 for the methanol production (de Riva, J., Ferro, V., Moya, C., Stadtherr, M.A., Brennecke, J.F., and Palomar, J. 2018). The process will be blacked box as the development cycle will be more focused on the CO_2 to Olefins and Olefins to monomers. The CO_2 Carbon Capture process is considered to be a backup as it is assumed that Marathon Galveston Oil Refinery has existing CO_2 Carbon capture to meet Environmental protection requirements.

To acquire the flue gas needed to extract the CO_2 for monomer production, the location of Galveston Oil refinery is selected for the burn stacks that release flue gas from methane burning. A fan will be placed within the top of the stack to push flue gas into a system of pipes where it will undergo the CO_2 processing. The contiguous operation of the plant will provide a steady amount of flue gas preventing disruptions and complications if the source stopped. Flue gas from methane is cleaner than most sources considered in the alternative process. The concept of removing flue gas from burn stacks helps reduce the amount of CO_2 in the atmosphere. The gas has 9.5% of CO_2 that can be removed and used in the MTO process.

The flue gas purification before reaching the original decided process are theoretical designs to apply to the concept of our future as deliberation with mentors lead to consensus of black boxing the two flue gas treatment steps to ease requirements for methanol production for olefins. More focus is needed in the development of Co2 to Methanol, Methanol to Olefins as the CO_2 carbon capture is standard process next to oil refineries, (Perry, Green, and O. Maloney 1997)

7.1.1.2 Operating Conditions

The operating conditions for extracting CO_2 from flue gas are to ensure that the CO_2 comes out in the best condition and to meet the 33,000 tones per year of CO_2 demand. This must be done through continuous operation throughout the year. Flue gas comes in as 9.5% CO_2. At proper operational conditions CO_2 purity from extraction is 90%. Conditions for equipment are for the absorber must operates at atmospheric pressure (1.00 bar) with top column temperatures of 40–50 °C, The stripper operates at 14.5 psia, with reboiler temperatures of about 140 °C and overhead temperatures of 60 to 95 °C, which producing 4000 kmol/hr of CO_2. After condensation step, the wet CO_2 product exits at 30 °C and about 1.00 bar with a typical composition of 80 mol% CO_2 and 20 mol% water. (CO_2 Utilization for Methanol Production, 2021)

Flue gas from Methane composition:

CO_2 (%vol)	H_2O (%vol)	O_2 (% vol)	N_2 (% vol)	CO (% vol)	NO_x (ppm)
12.0	**16.0**	**0.0**	**72.0**	**0.5**	**100**

7.1.1.3 Alternatives

An alternative method to extract flue gas for CO2 extraction was replacing the supply of steel mills with coal power plant flue gas. The constant emission of flue gas from burning presented a viable option due to the large number of emissions of flue gas. Using green methods to extract CO2 from the flue gas to repurpose it in ecological friendly uses was determined to be too costly as the CO2 purity in flue gas from coal is much contaminated with more dust than the flue gas from flue plants. Complicating the process and requirement for more solid treatment to be able to be used in olefin production.

7.1.2 CO_2 to Methanol Process

7.1.2.1 Description

The chosen process converts carbon dioxide (CO_2) and hydrogen (H_2) into methanol (CH_3OH) through catalytic hydrogenation. Four major steps take place: Feed Preparation, Methanol Synthesis, Separation, and Purification.

The process begins by preparing and conditioning the feed streams before they enter the catalytic reactor. Carbon dioxide and hydrogen are supplied as different feeds and are initially compressed to the required reaction pressure. They are then mixed in mixing units MIX-200 and MIX-201. This combined stream forms the stoichiometric CO_2/H_2 ratio necessary for methanol synthesis. The stream is then routed through a heat exchanger that recovers heat from hot process streams. Exchanges in E-201 elevate the temperature of the feed, reducing the amount of energy required later in the heater. Finally, an external heater, H-200, raises the stream temperature to approximately 210°C, ensuring that the mixture reaches the reactor at optimal reaction conditions.

One conditioned, the feed enters the methanol synthesis reactor (R-200), a catalytic reactor packed with $Cu/ZnO/Al_2O_3$ catalyst. This catalyst system is widely used in industrial methanol production and is well supported by various studies (Yusuf & Almomani, 2023). Within the reactor, CO_2 and H_2 undergo exothermic hydrogenation reactions to form methanol and water. The primary reactions are the following:

$$CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$$

$$CO + 2H_2 \rightarrow CH_3OH$$

In addition, the reverse water-gas shift (RWGS) reaction also takes place:

$$CO_2 + H_2 \rightarrow CO + H_2O$$

Although RWGS introduces an additional reaction pathway, CO is also consumed in methanol formation, and kinetic studies have shown that CO_2 remains the main carbon source for methanol synthesis under these conditions. The reactor operates at high pressure and moderate temperature to maximize conversion, following the design conditions described by Yusuf et al. (Yusuf & Almomani, 2023).

The hot reactor effluent exits R-200 and immediately enters a cooling and condensation train. Heat exchangers E-201, E-202, E-203, and E-205 remove heat from the rection products, lowering the temperature enough to condense crude methanol and water from the gas mixture. The cooled two-phase stream flows into the first separation drum, D-200, where liquids and gases are separated. The liquid contains crude methanol, water, dissolved CO_2, and trace components, while the vapor stream consists primarily of unreacted CO_2, H_2, and small amounts of CO. To improve overall conversion, the vapor stream is recycled back to the feed system. A small purge is taken to prevent the accumulation of inerts (Stream 14).

The crude liquid from D-200 undergoes additional cooling and flashing through exchanger E-204, and separation drum D-201 to remove remaining light gases. After gas removal, the liquid is preheated in E-205 and E-206 and directed to the purification section. Distillation column T-200 separates the methanol-water mixture into high purity methanol and water-rich bottoms stream. The overhead vapor is condensed in E-208 and collected in D-203 as the final methanol product. Overall, the process integrates compression, heat recovery, catalytic reaction, and separation into a continuous process that converts CO_2 to methanol to achieve a methanol purity of approximately 99%.

7.1.2.2 Operating Conditions

The operating conditions for the CO_2 to methanol process were selected based on published kinetic and process studies to achieve high conversion and efficient system performance (Yusuf & Almomani, 2023).The CO_2 feed enters at 12.3°C and 47.63 bar, while hydrogen enters at 25°C and 30 bar. Both streams undergo compression so that the final mixed feed reaches a pressure of approximately 75-78 bar before entering the reactor. Operating at this elevated pressure is essential because the methanol synthesis reactions involve a decrease in volume, and high pressure shifts the equilibrium toward methanol formation. Before entering the reactor, the feed mixture is heated to around 210°C using a combination of heat recovery exchangers and heaters. This temperature is consistent with the optimal range for $Cu/ZnO/Al_2O_3$ catalysts, which show strong activity between 200 and 260°C (Yusuf & Almomani, 2023).

Inside the reactor, the system operates at an inlet pressure of 75.7 bar and a controlled temperature of 210°C. These values match the specifications from the referenced study, which demonstrated that these conditions yield approximately 96% CO_2 conversion and 99% methanol production under isothermal operation (Yusuf & Almomani, 2023). The reactor contains a large mass of $Cu/ZnO/Al_2O_3$ catalyst (44,500 kg), and a pressure drop of about 0.6 bar occurs across the reactor.

Following the reaction, the effluent enters the separation and purification stages. The first drum operates near reactor pressure (approximately 73-74 bar) to condense crude methanol without excessive vaporization. Subsequent flash drums operate at reduced pressures of 1-2 bar to remove dissolved gases and prepare the liquid mixture for distillation. The methanol purification column operates close to atmospheric pressure with a reflux ratio of 2.12. Temperatures within the column typically range between 64°C and 102°C, depending on stage composition. The condenser cools the overhead product to around 14.95°C, allowing for the collection of high purity methanol. These operating conditions allow the process to achieve stable operation, high conversion, and efficient energy use.

7.1.2.3 Alternatives

Although the original process uses the $Cu/ZnO/Al_2O_3$ catalyst, several alternative catalyst systems have been proposed to improve performance in CO_2 hydrogenation. The standard $Cu/ZnO/Al_2O_3$ catalyst offers high activity and good selectivity at moderate temperatures. However, it is sensitive to temperature fluctuations and can deactivate through copper sintering under certain conditions (Tedeeva, 2024).

One potential alternative is the $Cu/ZnO/ZrO_2$ catalyst, where zirconia is added to increase CO_2 absorption and stabilize copper dispersion. Studies show that ZrO_2 can create stronger metal-support interactions and additional active sites, leading to improved methanol yields.

Another alternative includes oxide-based catalysts such as In_2O_3 or In_2O_3-ZrO_2. These materials activate CO_2 through oxygen vacancy sites rather than copper surfaces, which can enhance methanol selectivity and potentially reduce the influence of side reactions (Tedeeva, 2024). However, indium is more expensive than copper, and large-scale validation is still needed to confirm durability and performance under typical methanol synthesis conditions.

Overall, while these alternatives show potential, they remain less employed than the $Cu/ZnO/Al_2O_3$ system. Given its strong industrial track record and predictable behavior under high-pressure operation, this catalyst continues to be the most practical aand reliable choice for this process.

7.1.3 Methanol to Monomer Process

7.1.3.1 Description

The Methanol-to-Olefins (MTO) process converts high-purity methanol into light olefins, primarily ethylene (C_2H_4) and propylene (C_3H_6), which are essential building blocks for polymers such as polyethylene and polypropylene. This process serves as an alternative to traditional olefin production from fossil feedstock, such as steam cracking of naphtha or ethane. This enables the utilization of methanol derived from sustainable or alternative carbon sources such as captured CO_2.

The MTO Reaction proceeds through an intermediate dimethyl ether (DME) formation step, followed by conversion over a shape-selective zeolite catalyst, which in this case is SAPO-34. The whole mechanism encompasses numerous intermediates, which comprise two general pathways within the hydrocarbon pool theory: the olefin cycle and the aromatic cycle. The olefin cycle is direct oligomerization and β-scission of light olefins, specifically enhancing propylene generation. The aromatic cycle is the formation of aromatic intermediates that undergo chain growth and cracking to produce both ethylene and propylene.

These reactions are strongly influenced by the catalyst's pore architecture, acidity, and diffusion constraints, which are intentionally engineered to maximize light-olefin selectivity. SAPO-34 molecular sieve catalysts are widely used in commercial MTO units because their chabazite (CHA) framework provides small cages that preferentially crack intermediates to $C_2 – C_3$ olefins rather than heavier hydrocarbons (Ivar M, 1993),

Figure 6: Chemical Reactions of MTO Process *(Ivar M, 1993)*

1. Methanol Dehydration: formation of DME
$$2CH_3OH \rightarrow CH_3OCH_3 + H_2O$$
2. Hydrocarbon Pool Mechanism
   a. Olefin-Based Cycle:
$$C_2H_4 + {CH_3}^+ \rightarrow {C_3H_7}^+ \rightarrow C_3H_6 + H^+$$
$$C_3H_6 + {CH_3}^+ \rightarrow {C_4H_9}^+ \rightarrow C_4H_8 + H^+$$
$$C_4H_8 \rightarrow C_2H_4 + C_2H_4$$
   b. Aromatic (Methylated-Benzene) Cycle:
$$CH_3OCH_3 + H^+ \rightarrow {CH_3}^+ + CH_3OH$$
$$C_6H_6 + {CH_3}^+ \rightarrow {C_7H_9}^+ \rightarrow methylbenzenes$$
$$methylbenzene \rightarrow C_2H_4 + C_3H_6 + (aromatics)$$
3. Side-Reactions:
   a. Heavy Olefins Through Oligomerization
$$C_3H_6 + C_3H_6 \rightarrow C_3H_{12}$$
   b. Aromatics via Cyclization and Hydrogen Transfer:

$$C_6H_{12} \rightarrow C_6H_6 + 3H_2$$

c. Coke Formation from Polyaromatics

$$Polyaromatics \rightarrow Coke + H_2$$

4. Olefin Formation (Simplified Overall Reaction):

$$C_2H_6O \rightarrow C_2H_4 + H_2O$$

$$3CH_3OH \rightarrow C_3H_6 + 3H_2O$$

The conversion of methanol results in coke formation on the catalyst surface due to condensation and polymerization of heavier olefins. Accumulated coke reduces catalyst acidity and pore accessibility, eventually lowering olefin yield. To maintain continuous high activity, MTO systems employ a circulating fluidized-bed reactor-regenerator configuration, modeled after the fluid catalytic cracking process. In this process, fresh, hot, regenerated catalysts are continuously fed to the reactor riser and fluidized by the incoming vaporized methanol. As coke is deposited during the reaction, the spent catalyst flows to a regenerator where the coke is combusted with air. The regenerated catalyst is returned to the reactor, maintaining uninterrupted operation and a consistent activity profile.

The reactor effluent is a high-temperature mixture containing ethylene-propylene, heavy olefins and paraffins, steam and water, aromatics and oxygenates, light gases, and trace amounts of methanol and DME. To stabilize selectivity and prevent post-reactor reactions, the effluent is rapidly quenched. This, in turn, halts further conversion and protects downstream equipment. Vapor-liquid separation removes condensed water and heavier hydrocarbons for recovery or recycling, while the gas phase proceeds to fractionation.

To separate the desired products, propylene and ethylene, from the gas phase, a separation system is used. A compressor is used to ensure proper separation pressures for the water knock-out drum, which removes condensed aqueous phases. Acid-gas removal is used to eliminate CO_2 and residual catalysts. Then, the flow goes through a series of fractionation columns, including a demethanizer, deethanizer, depropanizer, and C_3 splitter. The demethanizer removes light gases, such as fuel. The deethanizer isolates the C_2 fraction. The C_2 splitter separates ethylene and ethane. The depropanizer separates the C_3 fractions. The C_3 splitter separates propylene and propane. Purified ethylene and propylene leave the system as primary salable commodities for polymerization or petrochemical processing.

MTO Technology has emerged as a key enabler, allowing process regions that lack cost competitiveness to utilize liquid petroleum feedstocks. It allows diversification of olefin production pathways, enhances energy security, and leverages carbon-recycling strategies when integrated with CO_2-based methanol synthesis.

7.1.3.2 Operating Conditions

The MTO process is operated under conditions optimized to maximize light-olefin yield while minimizing heavy hydrocarbon formation and excessive coke deposition. Prior to entering the reactor, methanol is vaporized and typically mixed with dilution steam or water to regulate hydrocarbon partial pressure and suppress secondary reactions. The vaporized feed stream is introduced to the circulating fluidized-bed reactor at temperatures typically in the range of 400–430 °C, allowing for the rapid onset of catalytic dehydration and hydrocarbon pool reactions. As the conversion proceeds exothermically, the reactor outlet temperature rises and is typically controlled within the range of 450–500 °C by adjusting the catalyst circulation rate and diluent flow.

Reactor pressure is usually maintained at slightly above atmospheric levels, generally 1–3 bar(g), to reduce compression requirements and favor the formation of light olefins. The fluidized catalyst bed ensures excellent gas–solid contact, short residence time, and uniform temperature distribution, all of which contribute to high propylene and ethylene selectivity. The ratio of steam to methanol is a crucial control parameter that varies depending on the commercial technology but typically ranges from 0.5 to 1.5 mol of steam per mol of methanol.

As coke accumulates on the catalyst during reaction, a portion of the spent catalyst is continuously withdrawn to a separate regenerator. The regenerator typically operates in the range of 650–700 °C, where controlled combustion of coke with air restores catalyst acidity and pore accessibility. Regenerator pressure is slightly higher than that of the reactor, promoting catalyst flow back into the reaction zone without relying on mechanical transport.

Downstream of the reactor, the high-temperature product stream undergoes immediate quenching to halt further cracking and stabilize olefin selectivity. The pressure at this stage remains near that of the reactor to minimize expansion losses and maintain adequate density for efficient separation. The cooled stream is then directed to multi-stage compression and fractional distillation systems, which operate at conditions tailored to achieve polymer-grade olefins, including deep refrigeration and high-pressure splitting in the ethylene and propylene purification trains.

Overall, the operating window for the MTO process is deliberately narrow and carefully controlled, as conversion, catalyst stability, and product selectivity are highly sensitive to temperature, partial pressure effects, and solids circulation rate. Effective heat management and catalyst regeneration are therefore fundamental to achieving continuous, optimal performance in commercial MTO units.

7.1.3.3 Alternatives

Some alternatives exist within the MTO process, particularly in the selection of catalysts and reactor configurations, both of which have a strong influence on olefin yield, catalyst lifetime, and overall operating efficiency.

One catalyst alternative to SAPO-34 is MAPO-18, a small pore silicoaluminophosphate material with AEI framework. MAPO-18 has slightly larger cage openings and a different acidity profile that can reduce the rate of coke formation and extend catalyst life. Studies show that MAPO-18 allows improved diffusion of intermediates and can shift the product distribution to higher propylene selectivity (Xie, 2022). In addition, modifying the framework composition through metal incorporation can tune the acidity and reduce deactivation (Xie, 2022), which is a major challenge for traditional SAPO-34. These properties make MAPO-18 an alternative catalyst for improving overall MTO performance.

Alongside catalyst alternatives, different reactor configurations can also be considered. Although the MTO process typically uses fluidized-bed reactors due to their ability to continuously regenerate catalyst and maintain stable temperatures, fixed-bed reactors could potentially be an alternative. Fixed-bed reactors are simpler to construct and have lower capital costs. However, they face challenges such as coke deposition and faster catalyst deactivation compared to fluidized systems (Yuan, 2017).

Overall, these alternatives demonstrate some flexibility within the MTO process. However, each alternative comes with trade-offs in catalyst lifetime, regeneration needs, and overall efficiency. Therefore, further research is needed to determine the combination of catalyst and reactor design that provides the best balance of performance and economic feasibility for the MTO process.

7.2 Competing Processes

7.2.1 Methanol to Gasoline

7.2.1.1 Description

The Methanol-to-Gasoline (MTG) process is a catalytic conversion method that produces high-octane gasoline from methanol. The process begins by vaporizing and pre-heating crude methanol, which is sent to a dimethyl ether (DME) reactor. In this reactor, ethanol is partially dehydrated over alumina catalyst to form a mixture of DME, methanol, and water. This reaction is slightly exothermic and reaches an equilibrium mixture at moderate temperatures.

Next, the DME-rich stream flows into the ZSM-5 catalyst reactors, where the major conversion to hydrocarbons occurs. ZSM-5 promotes a complex series of reactions that rearrange methanol and DME molecules into gasoline hydrocarbons. These products include normal paraffins, isoparaffins, aromatics, and small amounts of light gases. Most of the rection heat is released in this step.

The reactor effluent is then cooled and separated into gas, water, and hydrocarbon liquid. The gas stream (mostly hydrocarbons, CO and CO_2) is partially recycled to moderate reactor temperature and maintain stable operation. The water phase, which contains small, oxygenated compounds, is sent to biological treatment.

The hydrocarbon liquid, known as raw MTG gasoline, is distilled to remove light gases and undesirable components. One important treatment step is the heavy gasoline treating unit (HGT), which removes durene (1,2,4,5-tetramethylbenzene). After durene reduction, the treated gasoline is blended with other fractions to meet product specifications. A block flow diagram of the process is shown below.

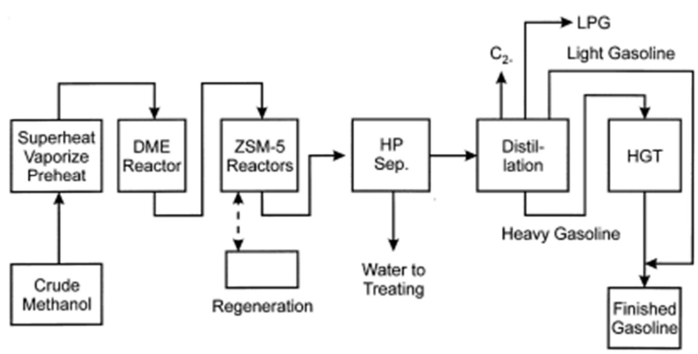


Figure 7: *MTG Block Flow Diagram (Keil,1999).*

Like any industrial process, MTG has both advantages and disadvantages. One major advantage is that the gasoline produced has high octane quality and can be used directly in existing engines. The process is also appealing because it can be made from non-petroleum derived sources, giving it potential environmental value. In contrast, some disadvantages relate to catalyst performance and product composition. The ZSM-5 catalyst gradually deactivates due to coke build-up, requiring periodic regeneration. Additionally, the process naturally forms a significant amount of durene, which must be removed to ensure acceptable fuel quality. The MTG reaction is also highly exothermic, so careful temperature management is required. Despite these challenges, the MTG process remains a proven route for producing gasoline from methanol.

7.2.1.2 Operating Conditions

The MTG process operates under well-defined conditions to ensure efficient conversion and stable catalyst performance. In the first stage, the DME rector typically runs at temperatures of about 310-320°C and a pressure close to 26 bar. These conditions allow methanol to reach its equilibrium mixture with dimethyl ether and water while releasing a modest portion of the overall process heat.

The second stage operates at higher temperatures of approximately 350-366°C and pressures of 19-23 bar. The ZSM-5 reactors are where most of the reaction heat is released, as the DME and methanol molecules are rearranged into gasoline hydrocarbons. To maintain stable operation, the reactors are typically arranged in parallel, with one reactor periodically being deactivated for catalyst regeneration. As the effluent exits the reactors, it is cooled to 25-35°C using cooling water before entering a three-phase separator, where gas, water, and liquid hydrocarbons are removed. Downstream distillation and treating steps then separate the gasoline fractions and remove unwanted components such as durene. These controlled operating conditions allow the MTG process to achieve high methanol conversion, consistent gasoline quality, and reliable long-term performance.

7.2.2 Ethanol to Ethylene

7.2.2.1 Description

The process converts ethanol into ethylene using a fluidized-bed dehydration reactor with catalyst regeneration. Ethanol vapor enters through the bottom of the fluidized-bed reactor, where it interacts with a catalyst for such alumina. The upward flow of ethanol maintains the catalyst in a fluidized state, ensuring excellent heat transfer and uniform temperature. Inside this reactor, ethanol is dehydrated into ethylene at high temperatures around 370 to 480 Celsius. This allows for very short processing times that last around 10 seconds to complete. The short processing time allows for a very high yield of 90% conversion.

The reactor waste product consists of ethylene and water; the waste products can be easily repeated downstream in the olefin production step. A portion of the catalyst continuously circulates out of the reactor as a way to mitigate the use of the new catalyst for lower cost and waste emissions. A regeneration step can be used to remove water from the spent catalyst. The regenerator consists of two streams used to pass hot air for drying and a vent and a second stream to push out humid air.

The high temperature from the streams that supply the reactor can be combined with cooler streams that require heating such as stream 2 and stream 4. To mitigate the use of energy in heating according to the patent concept of ethane to ethylene processing. Reactor flue gases can be cycled into the initial filtration step to decrease waste and increase CO_2 extraction in order to lower waste production exit through a vent line.

Overall, the system must operate continuously as the constant feed of ethanol leads to a constant amount of waste products that can be reincorporated into the overall system as a viable alternative to olefin production. The issue of running this with other parts of the olefin and CO2 production is with the amount of heat used to produce olefins required is the amount of heat needed to run an ethylene reaction is too high and must be contained.

7.2.2.2 Operating Conditions

To operate this plant, the ethanol dehydration reactor must operate at 345 to 480 °C. With vaporized ethanol contacting a fluidized bed of silica–alumina catalyst and a very short residence time of 1 to10 seconds. The endothermic reaction is supplied heat by circulating regenerated catalyst from a separate regenerator, which operates at 1540 to 705 °C with preheated air to burn off coke and restore catalyst activity. The regenerated catalyst returns hotter than the reactor to sustain reactor temperature, while flue gas leaves the regenerator slightly above atmospheric pressure. Overall ethanol exceeds 99%, while ethylene yields typically 95%.

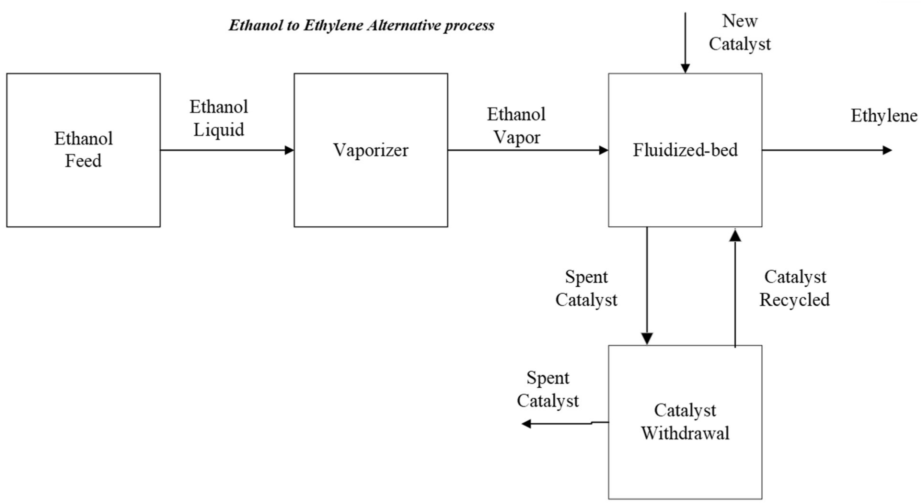


Figure 8: Ethanol to Ethylene Block Flow Diagram

7.2.3 CO_2 to Syngas Process

Due to the harmful effects of CO_2, many alternative projects exist that capture and utilize CO_2 in different ways. One alternative process involves converting CO_2 to synthesis gas (or syn gas) via RWGS using methane as a hydrogen and carbon source that's obtained using steam methane reforming (SMR). The heat source for the SMR and

RWGS reactors is generated through electrical energy, eliminating the need for hydrocarbon fuels.

The conversion of carbon dioxide to syngas through the reverse water-gas shift (RWGS) reaction is a multi-stage process that integrates chemical transformations with essential unit operations to achieve efficient CO_2 utilization. The process begins with CO_2 feed preparation, in which captured carbon dioxide, typically sourced from flue gas or direct air capture, is purified and compressed to the required operating conditions. In a parallel process, hydrogen production is carried out via water electrolysis, typically powered by renewable electricity, ensuring a sustainable supply. These two streams are introduced into the RWGS reactor, an electrified endothermal unit maintained at high temperatures (700–900°C) through resistive or "Joule" heating. Within this reactor, the primary reaction is the reverse water-gas shift, where CO_2 and H_2 react to form CO and H_2O. Secondary reactions may occur depending on operating conditions, including the forward water-gas shift ($CO + H_2O \rightarrow CO_2 + H_2$), which can partially equilibrate the system, and methanation side reactions ($CO + 3H_2 \rightarrow CH_4 + H_2O$), though these are minimized under RWGS conditions to preserve syngas selectivity.

Following the reactor stage, the process employs several unit operations to condition and separate products. Heat exchangers recover sensible heat from the hot effluent stream, improving overall energy efficiency. Condensers remove water vapor formed during the RWGS reaction, leaving a dry syngas mixture. Gas separators or membrane units may be used to adjust the H_2/CO ratio to meet downstream synthesis requirements, such as methanol or Fischer–Tropsch fuel production. In some configurations, acid gas removal (AGR) units are integrated to eliminate residual CO_2, ensuring a clean syngas stream. The final product is a tailored mixture of CO and H_2, ready for subsequent chemical synthesis.

This process flow demonstrates the interplay between chemical reactions and engineering operations. The RWGS reaction provides the fundamental transformation of CO_2 into CO, while unit operations such as compression, heating, condensation, and separation ensure that the syngas is produced efficiently, selectively, and in a form suitable for industrial application. The electrification of the reactor distinguishes this pathway from conventional thermal systems, offering precise temperature control, rapid response to fluctuating energy inputs, and compatibility with renewable electricity. Together, these features position the electrified RWGS process as a technically robust and environmentally favorable route for CO_2 utilization, contingent on continued advancements in catalyst stability, reactor design, and integration with renewable hydrogen supply chains.

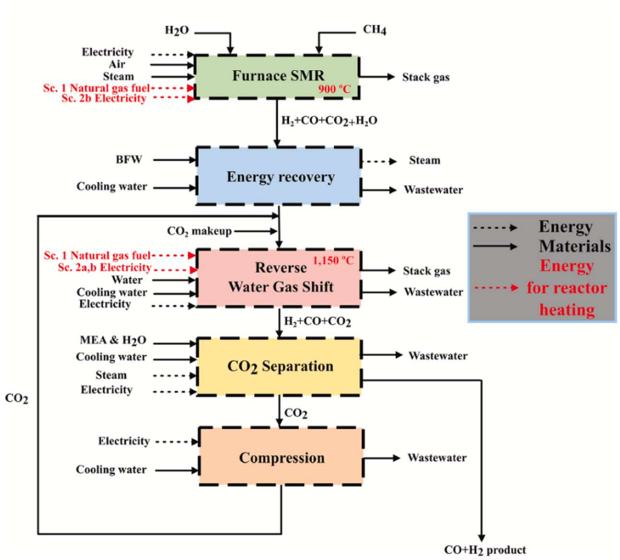


Figure 9. *SMR + RWGS for CO_2 to Syngas (Cao, 2022).*

8 Conclusion and Next Steps

This project shows how captured CO_2 from a refinery's methane burns tack flue gas can be converted into useful polymer products. By converting CO_2 into methanol and then into olefins, the process helps reduce industrial carbon emissions while also meeting the growing demand for plastics. Light olefins, such as ethylene and propylene, are used for the production of many everyday materials, and finding cleaner ways to produce them is becoming increasingly important as industries look for lower-carbon alternatives to traditional fossil-based routes.

However, producing only ethylene and propylene is not enough to maximize the value of this process. The market prices for polyethylene (PE) and polypropylene (PP) are significantly higher than the prices of their monomers. Because the suggested process requires a large amount of hydrogen, which has a high cost, it is necessary to create higher value final products to make the project financially feasible. Selling the monomers directly would not generate enough profit to balance out the high hydrogen costs, for which adding polymerization to the process is an essential step. By turning ethylene into PE and propylene into PP, the plant can sell products with much higher market value, which greatly improves the economic outlook of the entire system.

Even though this design can significantly reduce CO_2 emissions, further improvements are needed to make it commercially competitive. One of the biggest challenges is the cost of hydrogen. Future work should focus on finding ways to lower hydrogen production costs by integrating steam methane reforming (SMR) with CO_2 capture. Additional ways to reduce overall costs include improving heat integration to reduce energy consumption, optimizing reactor and separation performance, reducing the number of compressors being used, and extending catalyst life to minimize replacement costs.

Overall, this project shows that converting CO_2 into polymers would be a promising approach to lower carbon emissions in plastic manufacturing. With improvements that reduce production costs and increase overall efficiency, especially in hydrogen supply and energy use, the process has the potential to become both environmentally friendly and economically sustainable.

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10 Appendix

10.1 Design Basis

The objective of this design is to utilize post-combustion CO_2 emissions from refinery flue gas as a sustainable carbon feedstock to produce polymer-grade ethylene (99.9 mol%) and propylene (99.5 mol%). The process integrates post-combustion CO_2 capture using an amine absorption system, followed by hydrogen supply and conditioning for downstream reaction requirements. Captured CO_2 is converted to methanol through catalytic hydrogenation, and the methanol intermediate is subsequently transformed into light olefins via a methanol-to-olefins (MTO) reactor employing a SAPO-34 catalyst. The resulting olefin products are then purified to meet polymer-grade specifications. The facility is designed to operate continuously at large scale, with emphasis on maximizing CO_2 utilization, optimizing energy efficiency, ensuring economic feasibility, and minimizing overall environmental impact.

10.1.1 Plant Size

The proposed facility is designed for large-scale olefin production to meet industrial demand while achieving significant CO_2 reuse.

Table 3: Plant Size Specifications

Parameter	**Value**
Annual Operating Time	8000 h/y
Construction Period	2 years
Start-up Capacity	50% design capacity during 3rd year

Plant Life	20 years
Methanol Production (intermediate)	388,100 lb/hr
Ethylene Production	35,024 lb/hr
Propylene Production	35,024 lb/hr

10.1.2 Raw Material Requirements

The raw material supply is based on the existing oil refinery burn stack streams supplemented by necessary reaction inputs.

The amount of flue gas needed to operate plant is 7.32×10^9 m³/yr., which contains CO_2, N_2, O_2, H_2O, CO and NO_X .

Table 4: Composition of Flue Gas in vol%

Component	**Composition (vol%)**
CO_2	9.5 vol%
N_2	71.5 vol%
O_2	0.0 vol%
H_2O	19.0 vol%
CO	0.1 vol%
NO_X	50 ppm*

*Note: NO_x is stated in ppm due to small trace amounts of moles found in flue gas. Amount of the material is negligable to impede the process. (Engineers Notebook. Products of Combustion 2024)

The additional process inputs include water, steam, hydrogen gas, oxygen, and nitrogen. The consumables include the $Cu/ZnO/Al_2O_3$, SAPO-34, amine solvent, and nitrogen.

10.1.3 Location

The selected plant location is Houston, Texas, situated within the U.S. Gulf Coast refining and petrochemical corridor. This site was chosen due to its proximity to large refinery flue gas sources, ensuring a consistent and reliable supply of CO_2 for capture and conversion. The region also provides access to established hydrogen infrastructure and industrial utilities necessary to support methanol synthesis and MTO operations. Additionally, Houston's close proximity to polymerization facilities enables direct sales of polymer-grade ethylene and propylene, reducing transportation costs and logistical complexity. The area is supported by extensive port, pipeline, and rail infrastructure, facilitating efficient distribution of products and raw materials. Furthermore, federal incentives promoting carbon capture and utilization (CCUS) strengthen the economic viability of the project. Overall, Houston's highly integrated petrochemical ecosystem enhances supply chain efficiency and improves the overall feasibility of producing polymer-grade olefins from captured CO_2.

10.2 Block Flow Diagram

10.2.1 Current Process

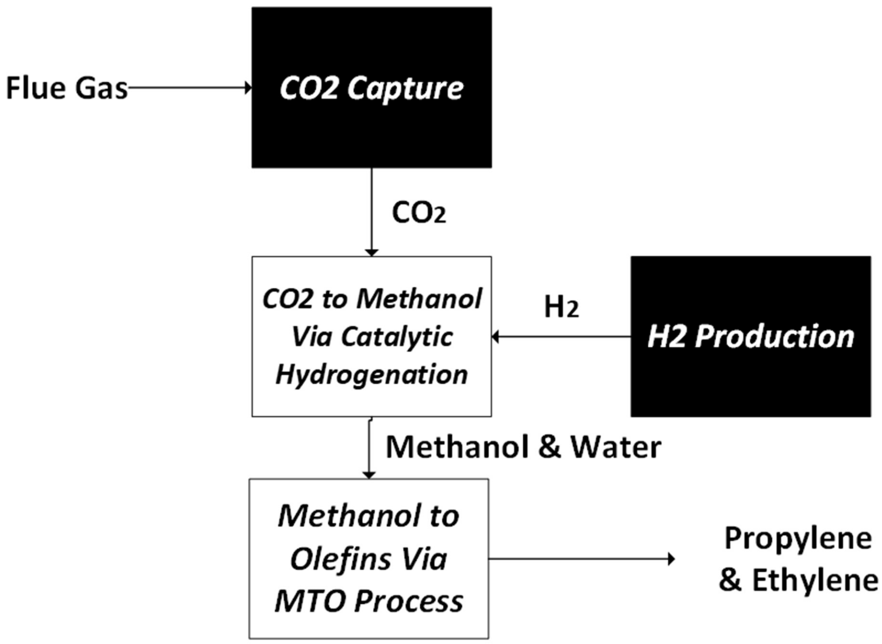
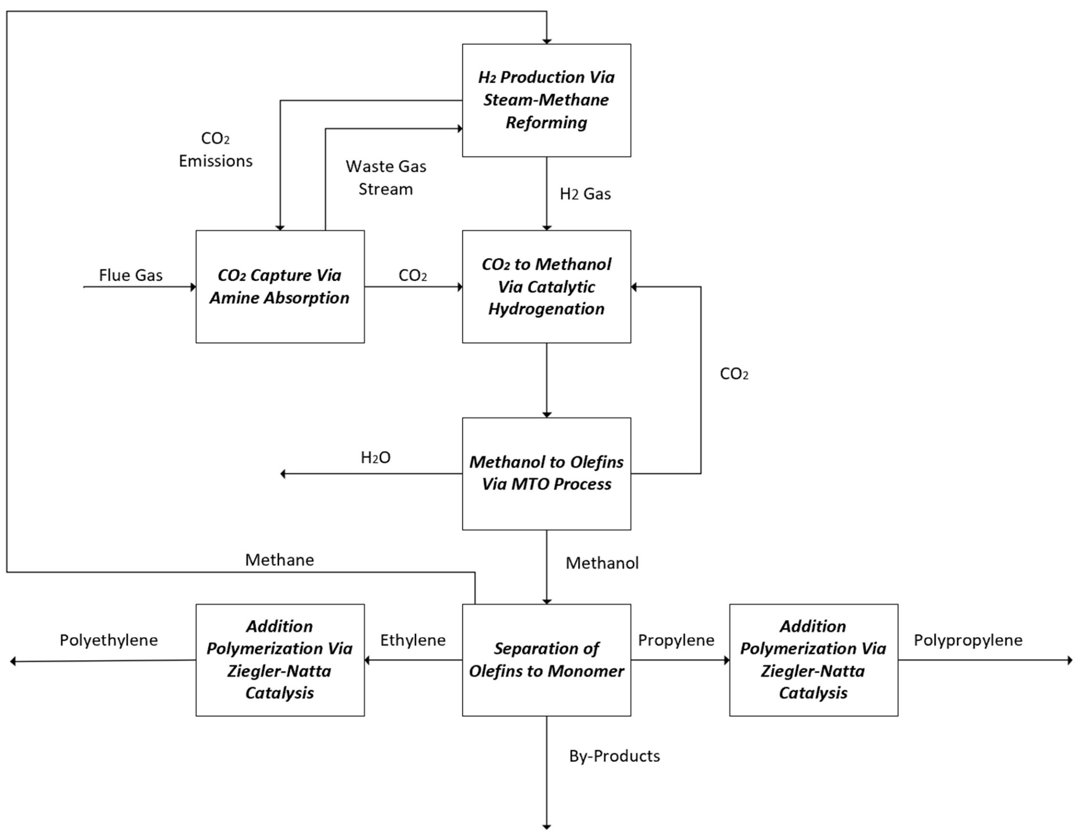


Figure 10: BFD of the Current Process

10.2.2 Ideal Case Process



10.3 Preliminary Process Flow Diagram

10.3.1 CO_2 to Methanol Process

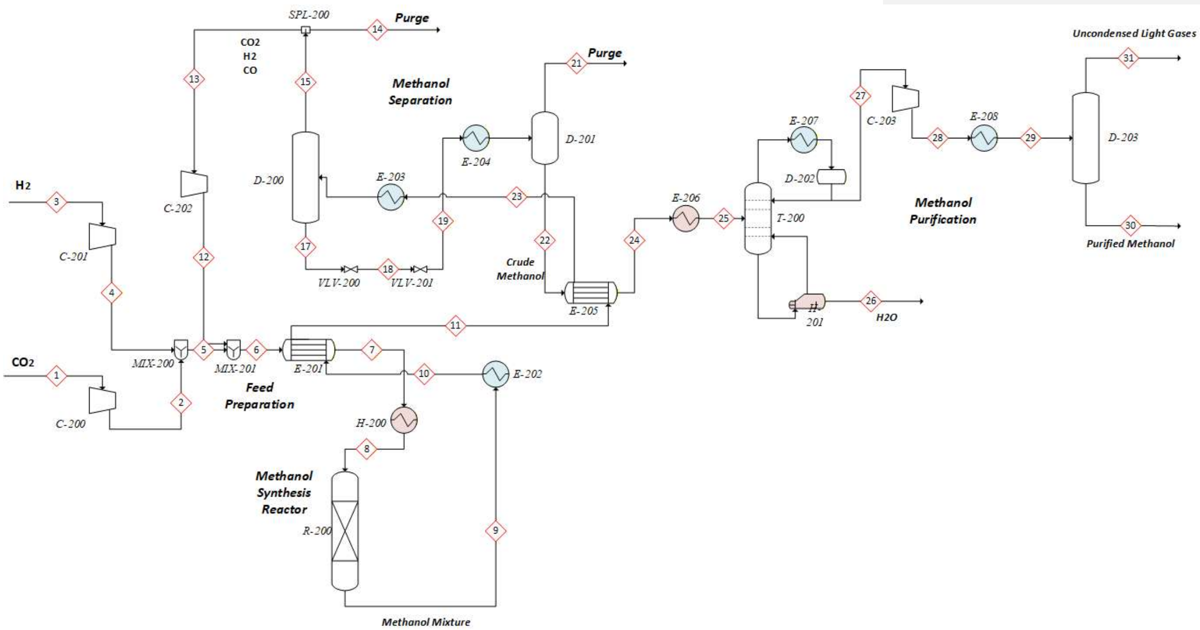


Figure 112: Reaction Section for CO2 to Methanol Process

10.3.2 Methanol to Monomer Process

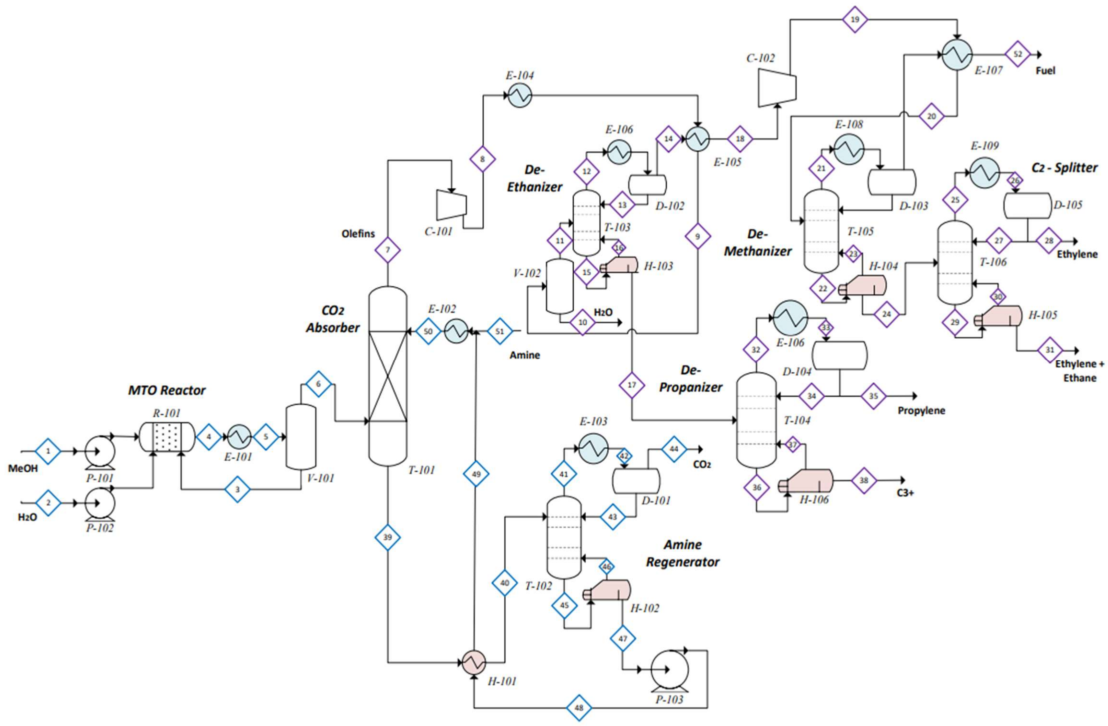


Figure 13: Methanol to Monomer Process Flow Diagram

10.4 Rough Material Balance

Commented [HN2]: Weight %

Table 5: Rough Material Balance of Methanol to Monomer Process

10.4.1 CO_2 to Methanol Process

Table 6: Rough Material Balance of CO_2 to Methanol Process

Stream	CO2	H2	MeOH	Water	CO	N2
1	1					
2	1					
3		1				
4		1				
5	0.74	0.260				
6	0.74	0.260				
7	0.74	0.239				0.021
8	0.74	0.239				0.021
9	0.028	0.051	0.508	0.343	0.071	
10	0.028	0.051	0.508	0.343	0.071	
11	0.028	0.051	0.508	0.343	0.071	
12	0.207	0.133			0.396	0.264
13	0.207	0.133			0.396	0.264
14	0.207	0.133			0.396	0.264
15	0.207	0.133			0.396	0.264
16	0.207	0.133			0.396	0.264
17	0.207	0.133			0.396	0.264
18	0.207	0.133			0.396	0.264
19			0.569	0.431		
20			0.569	0.431		
21	0.207	0.133			0.396	0.264
22			0.569	0.431		
23			0.569	0.431		
24			0.569	0.431		
25			0.569	0.431		
26			0.001	0.999		
27	0.057	0.066			0.42	0.457
28	0.057	0.066			0.42	0.457
29			0.569	0.431		
30			0.999	0.001		
31	0.057	0.066			0.42	0.457

10.4.2 Methanol to Monomer Process

Table 7: Rough Material Balance of Methanol to Monomer Process

Stream	MeOH	Water	C2H4	C3H6	Fuel	C_3^+	CO2	Amine	C2H6
1	1.00								
2		1.00							
3	0.64	0.36							
4	0.10	0.10	0.20	0.20	0.10	0.10	0.10		0.10
5	0.10	0.10	0.20	0.20	0.10	0.10	0.10		0.10
6		0.05	0.30	0.25	0.10	0.10	0.10		0.10
7		0.10	0.30	0.30	0.10	0.10			0.10
8		0.10	0.30	0.30	0.10	0.10			0.10
9		0.10	0.30	0.30	0.10	0.10			0.10
10		1.00							
11			0.35	0.30	0.13	0.12			0.10
14			0.50		0.20				0.30
17				0.80		0.20			
18			0.50		0.20				0.30
19			0.50		0.20				0.30
20			0.50		0.20				0.30
24			0.90						0.10
28			1.00						
31			0.20						0.80
35				1.00					
38						1.00			
39							0.30	0.70	
40							0.30	0.70	
44							1.00		
47								1.00	
48								1.00	
49								1.00	
50								1.00	
51								1.00	
52					1.00				

10.5 Final PFD and Mass Balance

10.5.1 CO2 to Methanol Process

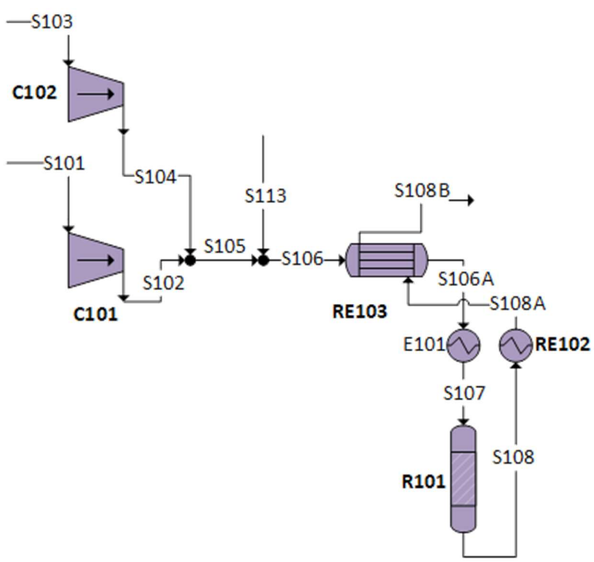


Figure 20: Reaction Section for CO2 to Methanol Process

Table 8: Mass Balance for Reaction Section of CO_2 to Methanol Process

Stream	Temp (°F)	Pressure (psia)	Phase	Mass Flow (lb/hr)	Mass Fraction				
					CO2	H2	MeOH	CO	H2O
CO2-S101	53.6	696.2	Vapor	388100	1	0	0	0	0
S102	127.4	1102.3	Vapor	388100	1	0	0	0	0
H2-S103	77.0	290.1	Vapor	124439	0	1	0	0	0
S104	424.4	1102.3	Vapor	124439	0	1	0	0	0
S105	352.4	1102.3	Vapor	512539	0.757	0.243	0	0	0
S106	170.6	1102.3	Vapor	1340000	0.441	0.488	0.026	0.042	0.003
S106A	239.0	1102.3	Vapor	1340000	0.441	0.488	0.026	0.042	0.003
S107	410.0	1102.3	Vapor	1340000	0.441	0.488	0.026	0.042	0.003
S108	498.2	1087.8	Vapor	1340000	0.158	0.45	0.23	0.044	0.119
S108A	374.0	1087.8	Vapor	1340000	0.158	0.45	0.23	0.044	0.119
S108B	303.8	1087.8	Vapor	1340000	0.158	0.45	0.23	0.044	0.119
S113	122.0	1102.3	Vapor	830437	0.245	0.639	0.043	0.068	0.005

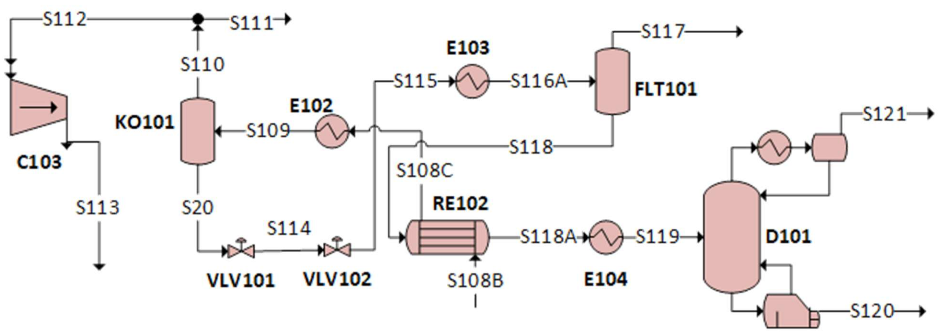


Figure 21: Purification Section for CO2 to Methanol Process

Table 9: Mass Balance for Purification Section of CO_2 to Methanol Process

Stream	Temp (°F)	Pressure (psia)	Phase	Mass Flow (lb/hr)	Mass Fraction				
					CO2	H2	MeOH	CO	H2O
S108C	303.8	1087.8	Vapor	1,340,000	0.158	0.450	0.230	0.044	0.119
S109	95.0	174.0	V + L	1,340,000	0.158	0.450	0.230	0.044	0.119
S110	113.0	1058.8	Vapor	916,165	0.231	0.659	0.041	0.064	0.005
S111	113.0	1058.8	Vapor	916,165	0.231	0.659	0.041	0.064	0.005
S112	113.0	1058.8	Vapor	830,437	0.245	0.639	0.043	0.068	0.005
S113	122.0	1102.3	Vapor	830,437	0.245	0.639	0.043	0.068	0.005
S114	116.6	609.2	V + L	427,959	0.002	0	0.635	0	0.363
S115	118.4	14.5	V + L	427,959	0.002	0	0.635	0	0.363
S116A	59.0	116.0	V + L	427,959	0.002	0	0.635	0	0.363
S117	59.0	14.5	Vapor	404	0.885	0.047	0.055	0.007	0.006
S118	59.0	14.5	Liquid	427,455	0.001	0	0.635	0	0.363
S118A	68.0	14.5	Liquid	427,455	0.001	0	0.635	0	0.363
S119	176.0	14.5	V + L	427,455	0.001	0	0.635	0	0.363
S120	195.8	14.5	Liquid	171,653	0	0	0.125	0	0.875
S20	113.0	1058.8	Liquid	427,959	0.002	0	0.635	0	0.363
S121	152.6	14.5	Vapor	256,000	0.002	0	0.977	0	0.020

10.5.2 Methanol to Olefins Process

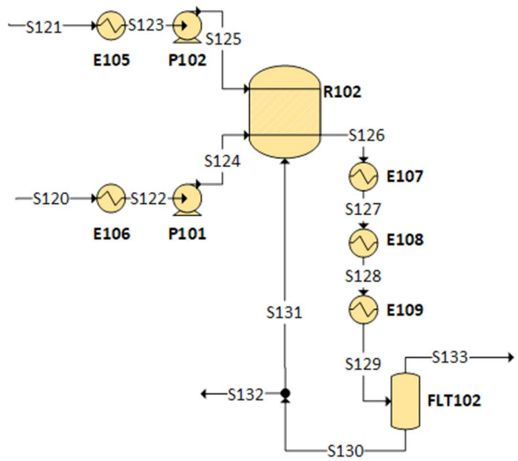


Figure 22: Absorber Section for Methanol to Olefins Process

Table 10: Mass Balance for Reaction Section for Methanol to Olefins Process

Stream	Temp. (°F)	Press. (psia)	Phase	Mass Flow (lb/hr)	Mass Fractions								
					CO2	H2	MeOH	C2H4	H2O	C3H6	C4H8	C5H10	CH4
S120	195.8	14.5	Liquid	171,163	0	0	0.125	0	0.875	0	0	0	0
S121	152.6	14.5	Vapor	255,801	0.002	0	0.978	0	0.020	0	0	0	0
S122	86.0	14.5	Liquid	171,653	0	0	0.125	0	0.875	0	0	0	0
S123	86.0	14.5	Liquid	255,801	0.002	0	0.978	0	0.020	0	0	0	0
S124	86.9	580.2	Liquid	171,653	0	0	0.125	0	0.875	0	0	0	0
S125	87.8	580.2	Liquid	255,801	0.002	0	0.977	0	0.020	0	0	0	0
S126	752.0	580.2	Vapor	1,650,000	0.002	0	0	0.0233	0.927	0.034	0.012	0.002	0
S127	392.0	572.9	V + L	1,650,000	0.002	0	0	0.023	0.927	0.034	0.012	0.002	0
S128	212.0	572.9	V + L	1,650,000	0.002	0	0	0.023	0.927	0.034	0.012	0.002	0
S129	123.8	572.9	V + L	1,650,000	0	0	0	0.023	0.927	0.034	0.012	0.002	0
S130	123.8	572.9	Liquid	1,530,000	0	0	0.001	0	0.999	0	0	0	0
S131	123.8	565.6	Liquid	1,220,000	0	0	0.001	0	0.999	0	0	0	0
S133	123.8	565.6	Vapor	121,278	0.02	0.001	0	0.318	0.003	0.463	0.16	0.029	0.006

Figure 14: Reaction Section for Methanol to Olefins Process

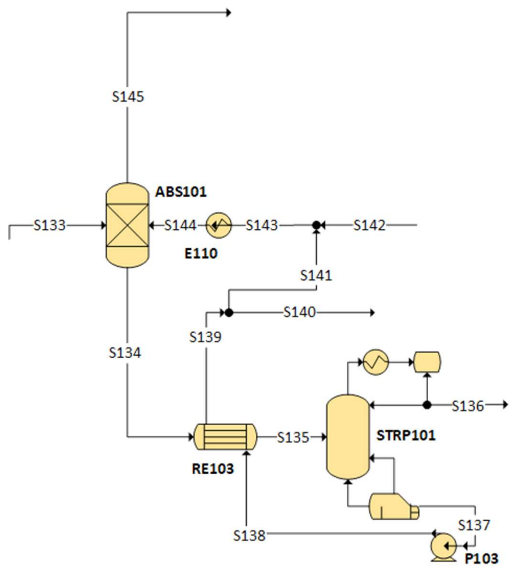


Figure 23: Absorber Section for Methanol to Olefins Process

Table 11: Mass Balance for Absorber Section for Methanol to Olefins Process

Stream	Temp. (°F)	Press. (psia)	Phase	Mass Flow (lb/hr)	Mass Fractions								
					CO2	H2	C2H4	H2O	C3H6	C4H8	C5H10	CH4	MDEA
S133	123.8	565.6	Vapor	121,278	0.02	0.001	0.318	0.003	0.463	0.16	0.029	0.006	0
S134	105.8	565.6	Liquid	607,772	0.004	0	0	0.131	0	0	0	0	0
S135	140.0	565.6	Liquid	607,772	0.004	0	0	0.131	0	0	0	0	0
S136	140.0	565.6	V + L	2,606	0.938	0	0.001	0.031	0.021	0.007	0.001	0	0
S137	178.6	10.9	Liquid	605,167	0	0	0	0.132	0	0	0	0	0.868
S138	181.4	580.2	Liquid	605,167	0	0	0	0.132	0	0	0	0	0.868
S139	149.0	580.2	Liquid	605,167	0	0	0	0.132	0	0	0	0	0.868
S140	149.0	580.2	Liquid	302,583	0	0	0	0.132	0	0	0	0	0.868
S141	149.0	580.2	Liquid	302,583	0	0	0	0.132	0	0	0	0	0.868
S142	104.0	580.2	Liquid	302,428	0	0	0	0.132	0	0	0	0	0.868
S143	126.5	580.2	Liquid	604,975	0	0	0	0.132	0	0	0	0	0.868

S144	86.0	580.2	Liquid	604,975	0	0	0	0.132	0	0	0	0	0.868
S145	123.8	152.3	Vapor	118,480	0	0.001	0.325	0.001	0.473	0.163	0.030	0.006	0

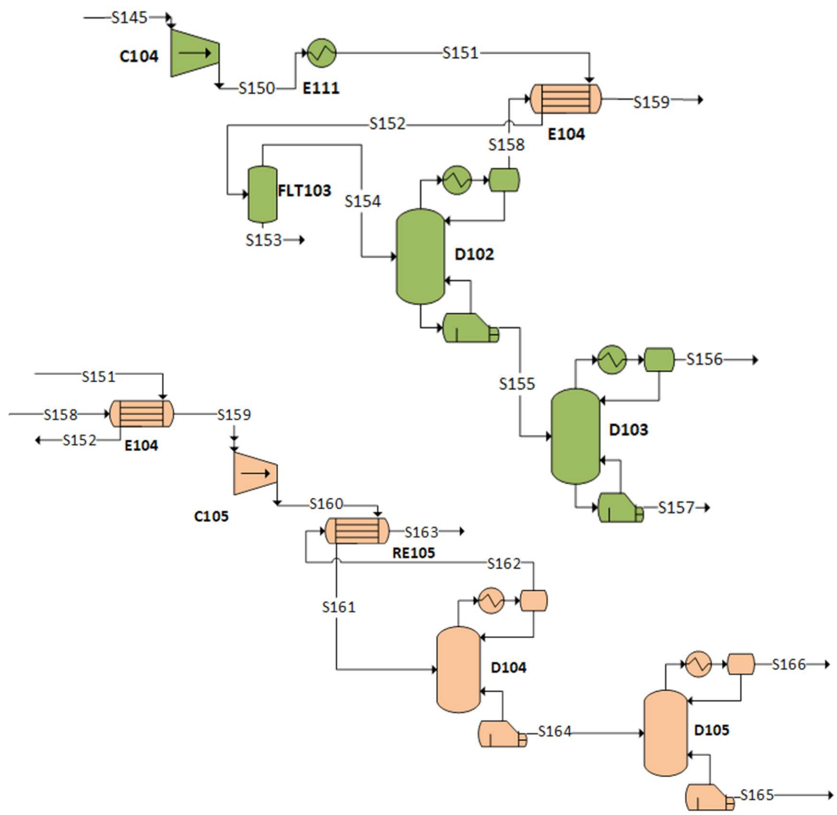


Figure 24: Purification Section for Methanol to Olefins Process

Table 12: Mass Balance for Purification Section for Methanol to Olefins Process											
Stream	**Temp (°F)**	**Press (psia)**	**Phase**	**Mass Flow (lb/hr)**	**Mass Fractions**						
					H2	**C2H4**	**H2O**	**C3H6**	**C4H8**	**C5H10**	**CH4**
S145	123.8	152.3	Vapor	118,480	0.001	0.325	0.001	0.473	0.163	0.030	0.006
S150	269.6	507.6	Vapor	118,480	0.001	0.325	0.001	0.473	0.163	0.030	0.006
S151	165.2	507.6	Vapor	118,480	0.001	0.325	0.001	0.473	0.163	0.030	0.006
S152	147.2	507.6	V + L	118,480	0.001	0.325	0.001	0.473	0.163	0.030	0.006
S154	140.0	456.9	Vapor	107,880	0.002	0.345	0.001	0.477	0.147	0.023	0.006
S155	195.1	514.9	Liquid	68,962	0	0	0.001	0.733	0.230	0.036	0
S156	119.5	290.1	Vapor	49,596	0	0	0	0.995	0.005	0	0
S157	245.5	290.1	Liquid	17,949	0	0	0.004	0.015	0.847	0.135	0
S158	-22.4	290.1	Vapor	38,917	0.004	0.955	0	0.023	0	0	0.017
S159	110.3	290.1	Vapor	38,917	0.004	0.955	0	0.023	0	0	0.017
S160	197.6	507.6	Vapor	38,917	0.004	0.955	0	0.023	0	0	0.017
S161	186.8	507.6	Vapor	38,917	0.004	0.955	0	0.023	0	0	0.017
S162	-72.4	493.1	Vapor	2,263	0.075	0.627	0	0.002	0	0	0.296
S163	59.0	493.1	Vapor	2,263	0.075	0.627	0	0.002	0	0	0.296
S164	18.5	493.1	Liquid	36,654	0	0.975	0	0.025	0	0	0
S165	134.8	362.6	Liquid	877	0	0.009	0	0.991	0	0	0
S166	-5.4	362.6	Vapor	35,024	0	0.999	0	0.001	0	0	0

10.6 Equipment Sizing

Table 13: Equipment List and Sizing					
Equipment	**Type**	**Material**	**Sizing**	**Duty (MMBtu/hr)**	**Work (hp)**
C101	Centrifugal Compressor	Carbon Steel	—	—	1,527
C102	Centrifugal Compressor	Carbon Steel	—	—	59,561
C103	Centrifugal Compressor	Carbon Steel	—	—	5,463
C104	Centrifugal Compressor	Carbon Steel	—	—	2,364
C105	Centrifugal Compressor	Carbon Steel	—	—	491

P101	Centrifugal Pump	Stainless Steel	—	—	193
P102	Centrifugal Pump	Stainless Steel	—	—	353
P103	Centrifugal Pump	Stainless Steel	—	—	632
R101	Fixed-Bed Plug Flow Reactor	Silicon Carbide	Diameter: 3 ft Length: 16 ft	0 (adiabatic)	—
R102	CSTR Reactor	Stainless Steel	Diameter: 11 ft Height: 40 ft	2,001	—
D101	Distillation Column	Stainless Steel	Diameter: 28.5 ft Height: 50 ft	Reboiler: 393 Condenser: -288.5	—
D102	Distillation Column	Stainless Steel	Diameter: 7.5 ft Height: 98 ft	Reboiler: 14.25 Condenser: -21	—
D103	Distillation Column	Stainless Steel	Diameter: 5 ft Height: 92 ft	Reboiler: 6 Condenser: -6	—
D104	Distillation Column	Stainless Steel	Diameter: 3 ft Height: 108 ft	Reboiler: 4 Condenser: -11	—
D105	Distillation Column	Stainless Steel	Diameter: 4 ft Height: 4 ft	Reboiler: 7 Condenser: -3	—
ABS101	Absorber	Stainless Steel	Diameter: 8.5 ft Height: 26.25 ft	—	—
STRIP101	Stripper	Stainless Steel	Diameter: 8.5 ft Height: 26.25 ft	14	—
KO101	Flash Drum	Stainless Steel	Diameter: 21 ft Height: 12 ft	0 (adiabatic)	—
FLT101	Flash Drum	Stainless Steel	Diameter: 8.5 ft	0 (adiabatic)	—

			Height: 25 ft		
FLT102	Flash Drum	Stainless Steel	Diameter: 11.5 ft Height: 36 ft	0 (adiabatic)	—
FLT103	Flash Drum	Stainless Steel	Diameter: 4 ft Height: 12 ft	0 (adiabatic)	—
E101	Steam Heater	Carbon Steel	Area: 35,930 sqft	422	—
E102	Cooling Heat Exchanger	Carbon Steel	Area: 40,591 sqft	-755	—
E103	Cooling Heat Exchanger	Carbon Steel	Area: 6,318 sqft	-25	—
E104	Heating Heat Exchanger	Stainless Steel	Area: 3,423 sqft	72	—
E105	Cooling Heat Exchanger	Carbon Steel	Area: 28,104 sqft	-150	—
E106	Cooling Heat Exchanger	Carbon Steel	Area: 2,368 sqft	-21	—
E107	Cooling Heat Exchanger	Carbon Steel	Area: 24,283 sqft	-1,572	—
E108	Cooling Heat Exchanger	Carbon Steel	Area: 12,228 sqft	-383	—
E109	Cooling Heat Exchanger	Carbon Steel	Area: 12,971 sqft	-165	—
E110	Cooling Heat Exchanger	Carbon Steel	Area: 5,920 sqft	-14	—
E111	Heating Heat Exchanger	Stainless Steel	Area: 377 sqft	-7	—
RE101	Heat Exchanger	Carbon Steel	Area: 11,184 sqft	-300	—
RE102	Heat Exchanger	Carbon Steel	Area: 108 sqft	4	—

RE103	Heat Exchanger	Carbon Steel	Area: 8,288 sqft	166	—
RE104	Heat Exchanger	Carbon Steel	Area: 151 sqft	2	—
RE105	Cooling Heat Exchanger	Stainless Steel	Area: 108 sqft	0	—

10.7 Cost of Utilities

Table 14: Cost of Utilities per Unit		
Equipment	**Utility Type**	**Cost ($/hr)**
E102	Cooling Water	305.19
E105	Cooling Water	54.995
E106	Cooling Water	7.8375
E108	Cooling Water	140.38
E109	Cooling Water	60.453
E110	Cooling Water	5.2849
E111	Cooling Water	2.5183
E103	Refrigerated Water	77.5402
E107	Refrigerated Water	7224.77
EI04	Low Pressure Steam	583.35
EI01	High Pressure Steam	5112.97
CI01	Electricity	68.3375
CI02	Electricity	2664.88
CI03	Electricity	289.636
PI01	Electricity	8.47706
PI02	Electricity	15.5136
PI03	Electricity	28.2639
CI04	Electricity	107.227
CI05	Electricity	21.5212

A majority of the utility usage and demand of the plant was calculated using the Aspen Plus v14 simulator. The heat exchanger

10.8 Sample Hand Calculation

<u>Minimum Number of Stages:</u>

The minimum number of theoretical stages was calculated using the Fenske equation:

$$N_{min} = \frac{\ln\left(\frac{z_D}{1-z_D} \cdot \frac{1-z_B}{z_B}\right)}{\ln(\alpha)}$$

$$N_{min} = \frac{\ln\left(\frac{0.99}{0.01} \cdot \frac{0.93}{0.07}\right)}{\ln(4)} = 5.18$$

Minimum number of stages = 6 trays

Using McCabe–Thiele analysis for theoretical stages: **12 trays**

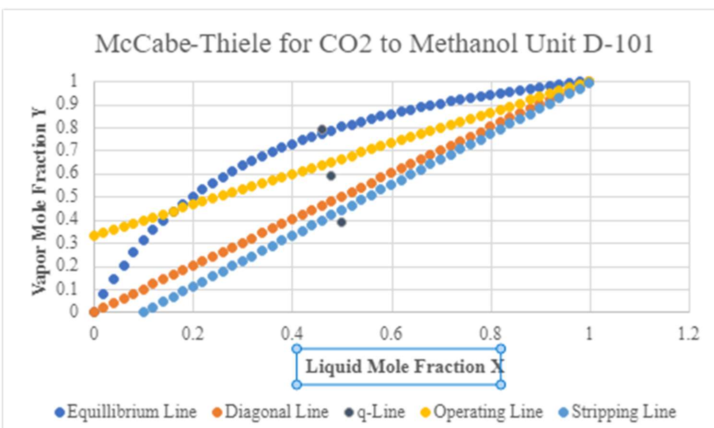


Figure 215: Absorber Section for Methanol to Olefins Process

Actual number of trays needed for Distillation Column (D-101):

$$E_0 = 0.70$$

$$N_{actual} = \frac{N_{min}}{E_0} = \frac{12}{0.70} = 17 \text{ trays}$$

Column Height

Tray spacing:

$$S = 2.0 \text{ ft}$$
$$H_{active} = N_{actual} \times S = 17 \times 2.0 = 34 \text{ ft}$$

Including reboiler and condenser allowances:

$$H_{total} = H_{active} + H_{reboiler} + H_{condenser}$$
$$H_{total} = 34 + 12 + 8 = 52.0 \text{ ft}$$

Total column height = 52.0 ft

Column Diameter

$$u_v = (-0.17L_t^2 + 0.27L_t - 0.047)\left(\frac{\rho_L - \rho_V}{\rho_V}\right)^{1/2}$$

$$u_v = (-0.17(0.61)^2 + 0.27(0.61) - 0.047)\left(\frac{790 - 8}{8}\right)^{1/2}$$
$$u_v = 0.83 \text{ m/s} = 2.71 \text{ ft/s}$$

Maximum vapor flowrate:

$$\dot{V} = (R + 1)D$$
$$\dot{V} = (2 + 1)(3531) = 10{,}593 \text{ kmol/hr}$$

Volumetric Flowrate

Using the ideal gas equation:

$$\dot{V} = \frac{nRT}{P}$$
$$\dot{V} = \frac{(10593)(8314)(353)}{1.23 \times 10^5}$$

$$\dot{V} = 70.2 \text{ m}^3/\text{hr} = 229.7 \text{ ft}^3/\text{s}$$

Diameter Calculation

$$D_c = \sqrt{\frac{4\dot{V}}{\pi u_v}}$$

$$D_c = \sqrt{\frac{4(70.2)}{\pi(0.96)}}$$

$$D_c = 9.63 \text{ m} = 31.6 \text{ ft}$$

Final Results

Table 15: Comparison of Hand Calculation and Aspen Results for D-101

Parameter	Hand Calculation	Aspen Results
Column Trays	17 trays	15 trays
Column Diameter	31.6 ft	28.5 ft
Column Height	52.0 ft	50.2 ft